Cross-Layer Dataset Design and Exploratory Analysis of an ESP32-Based ESP-WIFI-MESH Network

Thesis 3 | NIS16

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Overview

1 What we built

- 8 ESP32 boards: 1 root + 7 children
- Blackhole and wormhole, emulated at the application layer
- 4 topologies × 3 sites × 3 scenarios: 80 runs

2 What we checked

- A verifier tests every run before it enters the dataset
- Both attacks confirmed in 72 of 72 attack runs
- No attack signature in 8 of 8 benign runs

3 What we found

- Blackhole: visible through delivery (PDR)
- Wormhole: visible through duplicate arrivals at the root
- Radio and routing features did not move

Both attacks left a clear, verified mark on the traffic, but not on the radio or routing features, because they were emulated at the application layer.

1-3

Chapters 1 to 3

The gap and the objectives

1. **Simulated data:** Most IDS datasets come from simulators or IP networks (KDD, NSL-KDD), not from mesh hardware

2. **Missing layers:** Packet or flow datasets do not hold PHY, MAC and network signals of a microcontroller mesh together

3. **Real attacks:** Blackhole and wormhole detection is mature in simulation; on real ESP32 boards it is undocumented

This study: a labelled, cross-layer dataset from a real ESP-WIFI-MESH network under normal operation and two controlled attacks. Setting: campus monitoring, as in Khan et al. (ESP32 mesh, delivery above 97%).

General objective

Design, generate and analyse a cross-layer dataset from an ESP32 mesh to explore how normal operation differs from controlled attacks.

Specific objectives

1. Build an ESP32 mesh testbed with normal and attack traffic

2. Collect PHY, MAC and network features into a labelled dataset

3. Evaluate it with EDA and unsupervised clustering

Scope: a dataset and its exploratory analysis, not an intrusion detection system.

The theory: an attack breaks a relationship

Normal mesh traffic keeps cross-layer relationships (Table 3.7). We wrote down which ones each attack should break before any capture (Tables 3.4 and 3.5).

Relationship	Under normal operation	Prediction written before any data
Received ≈ forwarded	Every relay passes on what it gets	**Blackhole: ForwardingRatio < 1 at the attacker; victims' PDR falls; victims retransmit more**
One copy per probe	Each probe reaches the root once, by one path	**Wormhole: duplicate arrivals at the root**
Delay follows hops	Deeper nodes take longer to reach the root	**Wormhole: latency mismatch between the two copies**
Placement fits topology	Parent layer < child layer; signal weakens with depth	**Wormhole: low hop counts, impossible parent links, RSSI that does not fit the depth**

Chapter 6 tests every row. The tables were kept as written, so the panel can see what was predicted and what was found.

4-5

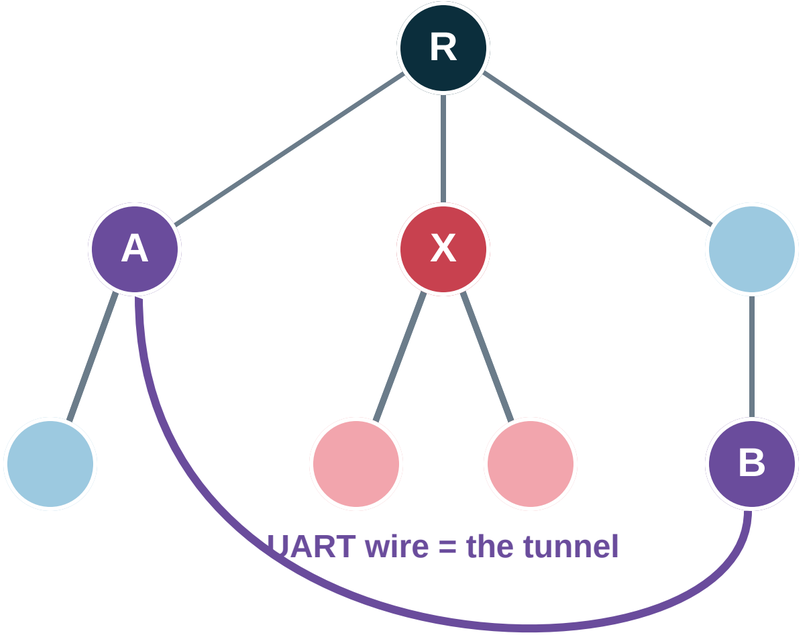
Chapters 4 and 5: planned vs. actual

How a run works, how both attacks are built, how every run is checked, and changed plans

NEW SLIDE

Our testbed and what it follows

WHAT WE BUILT



root · blackhole attacker · tunnel ends A, B · victims (behind X) · bystanders

8 ESP32 boards · 3 campus sites · 4 topologies · 3 scenarios

WHAT EACH PART FOLLOWS

Healthy ESP-MESH on ESP32 boards, campus monitoring setting	→	**Khan et al. [22]** — ESP-MESH on ESP32, campus test: delivery above 97%
Blackhole: a placed relay drops what it should forward	→	**Airehrour et al. [70]** — The blackhole definition we check against: 5 of 6 criteria met
Wormhole: a wire tunnels end B's probes to end A	→	**Zhukabayeva [71] · Hu [46] · Ramírez [72]** — Tunnel, definition (4 of 5 met), duplicate arrivals
Who is a victim: the boards behind the attacker	→	**Karlof and Wagner [81]** — The attacker's position decides the victims

No published blackhole or wormhole measurement exists on ESP-WIFI-MESH, so we compare direction and size, not exact values. Both attacks are emulated at the application layer.
Traffic is synthetic: each board sends a small test packet about once a second to the root, not real temperature or soil readings. Attacks act on forwarding, not content, and a known steady stream lets us count exactly what went missing (forwarding ratio, delivery ratio).

How one run works

Formation	Baseline	Attack	Cooldown	→ End
≥ 60 s	300 s	180 s	120 s	
label 255, excluded	label 0 (normal)	label 1 blackhole / 2 wormhole	label 0 (normal)	

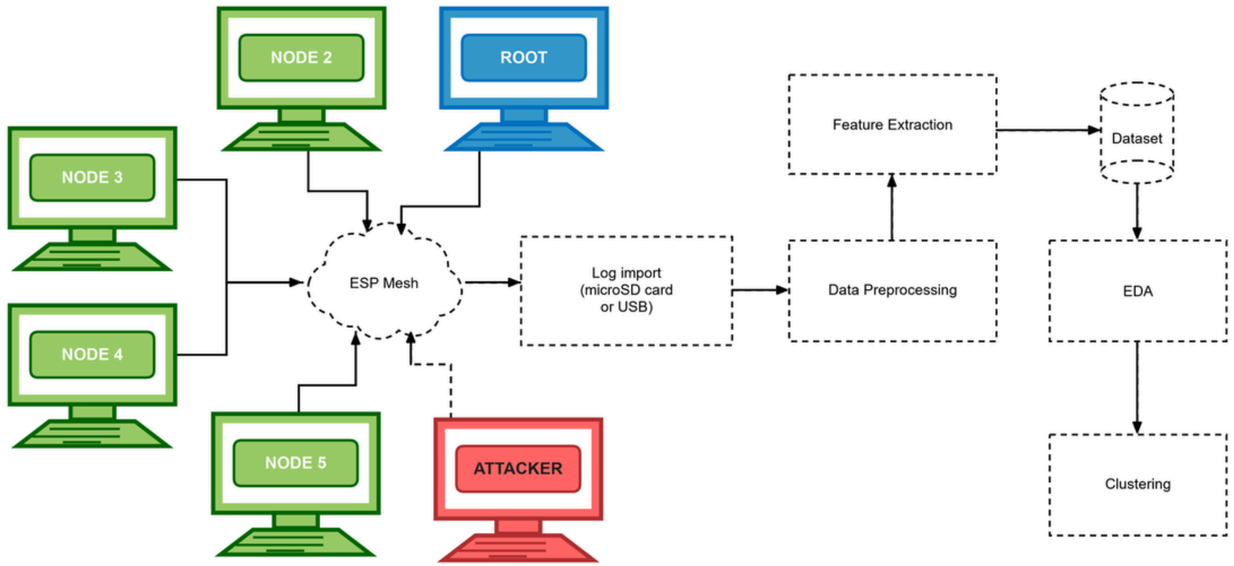


Figure 4.1. Read left to right: boards → ESP mesh → log import → preprocessing → features → dataset → EDA and clustering. Red board = attacker; dashed boxes = processing steps.

- Every board logs at 10 Hz to flash and a microSD card
- Logs are imported after the run; nothing is streamed over the mesh, so an attack cannot erase its own evidence
- The root drives the phases: 5 rounds, 100 ms apart, repeated every 10 s
- The attack starts only after the pre-run gates pass and the operator types START (D-22)
- Labels come from the phase schedule, never from a feature

Two attacks, emulated at the application layer

Blackhole: forwarding suppression

- A placed relay runs the same relay code as every node
- In the attack phase it stops forwarding (no esp_mesh_send() call)
- It still ACKs every frame at the link layer, so victims see no failure

✓ 5 of 6 criteria of Airehrour et al. [70] met

✗ **Not met: attracts traffic with a false route**

Wormhole: out-of-band tunnel

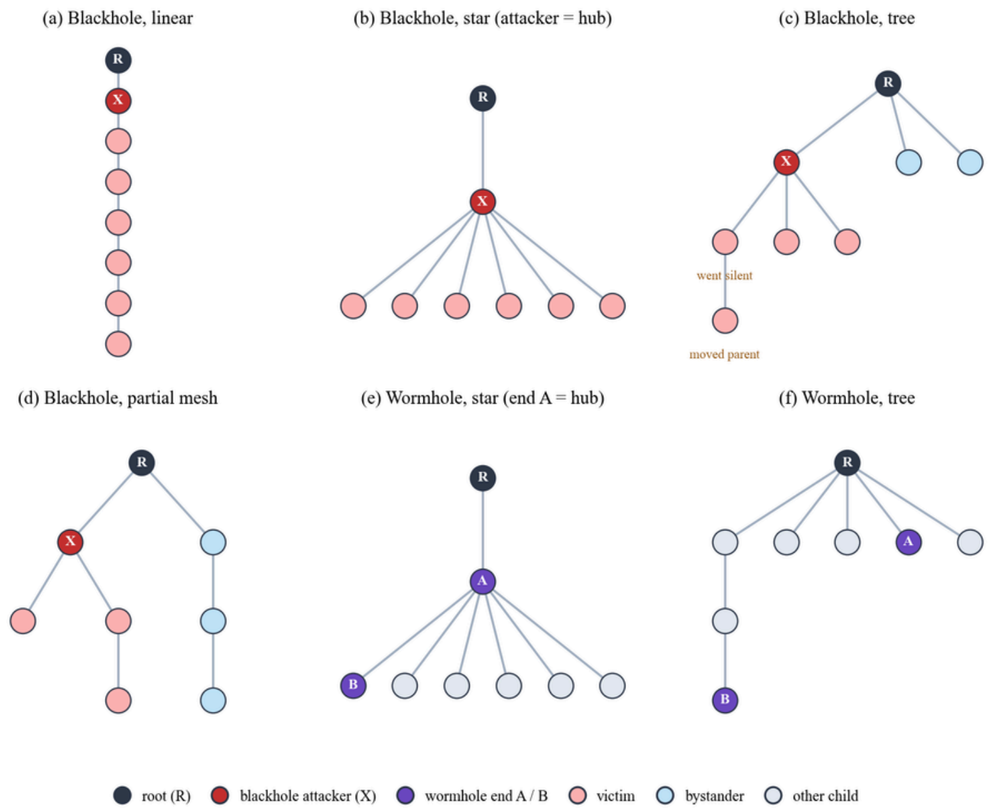
- UART wire, 115200 baud, 30-byte frames with CRC32
- End B (deeper) also writes its own probes into the wire; end A (shallower) re-injects them toward the root
- The root gets every B probe twice, by two paths, with different delays

✓ 4 of 5 criteria of Hu, Perrig and Johnson [46] met

✗ **Not met: creates a false neighbour**

Why emulation (D-20): ESP-WIFI-MESH lets an app read its routing table but not write it, so both attacks reproduce the traffic effect, not the routing manipulation.

Topologies as they formed, and the run design



Three sites

- G402 (GCCS): controlled classroom
- DLSU Library: busy area, heavy Wi-Fi
- Yuchengco Building: open area below the building (semi-outdoor)

Three traffic scenarios

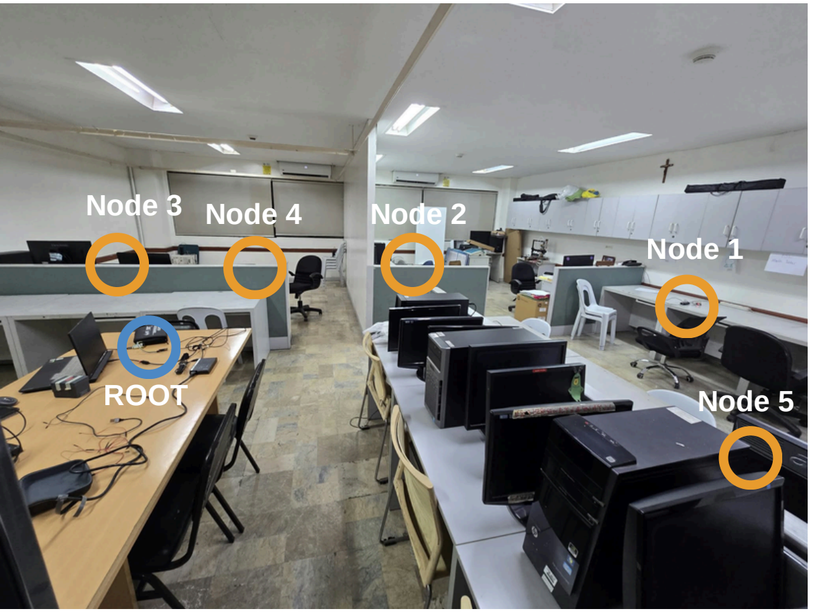
- Stationary: 1 probe per second (reference)
- Burst: 300 probes, 60 s into the attack
- Highload: 4 probes per second, whole run

4 topologies × 2 attacks × 3 sites × 3 scenarios = 72 attack runs + 8 benign runs = 80. One run per cell, so the sites act as repeats.

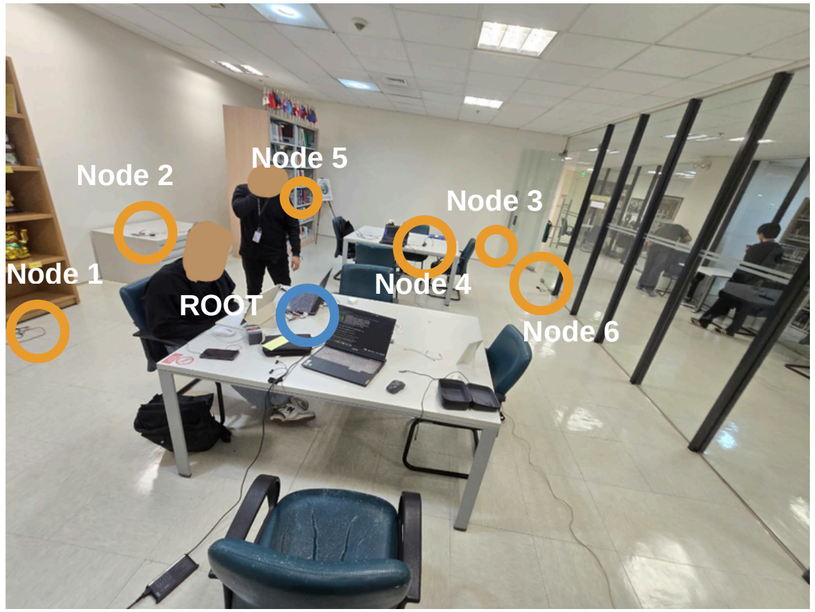
How to read it

Question: What shape did each network really take when the attack started?
How to read: Each drawing is one run's tree: root R on top, each line down is one hop. X = blackhole attacker, A / B = wormhole tunnel ends, pink = victims (behind X), light blue = bystanders, grey = other boards.
What you see: The star has no bystanders, because every board sits behind the hub. In (f), end B sits deeper than end A, so the tunnel skips hops.

Testbed Deployments Across Locations



G402



Library

Yuchengo

From raw rows to 16 features

1. **Align** — Boards share no clock; each is aligned to its own exit from phase 0
2. **Window** — 1 s windows on a 10 Hz grid; fewer than 8 of 10 samples = discarded
3. **Fill** — Gaps of 1–2 samples inside a kept window are filled
4. **Count** — Cumulative counters become per-window deltas
5. **Join** — Delivery joins root arrivals on the sequence number, never on time

One row = one board × one second

- **Forwarding**
  ForwardingRatio, IngressEgressDelta, ConsistencyScore
- **Delivery**
  PDR (blank when nothing was sent), RetryRate
- **Topology**
  ParentSwitchRate, HopChangeCount, HopStabilityDuration
- **Physical**
  RSSI_mean, RSSI_var, RSSI_stability
- **Cross-layer**
  RSSI_Hop_Diff, LatencyHopRatio
- **Tunnel**
  TunnelIntensity, TunnelBytes, TunnelLatency (ends only)

CHANGED

How to read the dataset: one row

One row = one board × one second. Four real rows from feature_table.csv of the tree / G402 / stationary blackhole run. Each header is the exact CSV column name, with its plain meaning underneath:

#	node_id last 4 hex	exposure role	segment phase	t_anchor_s s from attack	Label 1 = attack	hop 0 = root	recv_count_delta received	forward_count_delta forwarded	drop_count_delta dropped	ForwardingRatio fwd ÷ recv	PDR delivery	LatencyHopRatio relative delay
1	1C38	attacker	baseline	−100	0	1	3	3	0	**1.000**	blank	blank
2	1C38	attacker	attack	120	1	1	3	0	3	**0.000**	blank	blank
3	D7B4	downstream = victim	attack	120	1	2	0	0	0	blank	**0.000**	blank
4	0C80	upstream = bystander	attack	120	1	1	0	0	0	blank	**1.000**	6.161

1 Label = the phase

Rows 1 and 2 are the same board. Label flips from 0 to 1 only because segment went from baseline to attack (the phase schedule); no feature decided it.

2 Blank is not zero

Blank (an empty cell in the CSV) = can't judge. PDR is blank if the board sent nothing, was disconnected, or the root wasn't logging. Row 3's 0.000 is real: sent, none arrived.

3 Same label, different story

Rows 3 and 4 both carry Label 1, but the downstream victim's PDR is 0.000 and the upstream bystander's 1.000. That is why no method reaches ARI 1.0 on all nodes.

CHANGED

Every run is validated before it counts

1 Signature

3-sigma test (Zhukabayeva et al. [71]): attack mean ≥ 3 baseline SDs away in the expected direction, change ≥ 0.10

Studies: Z A R K

2 Topology

Tree rebuilt from parent addresses; it must match the topology we designed for the run

Studies: KW H

3 Dataset

Every board covers ≥ 95% of each phase; missing boards are graded by role

Our own rule, fixed in advance

4 Scenario

Probe rate and burst timing match the scenario the run is filed under

Our own rule, fixed in advance

Six published studies behind the checks

blackhole wormhole both

Z — Zhukabayeva et al. [71]
ZigBee testbed: normal vs. attack judged by a 3-sigma rule; the wormhole is an out-of-band tunnel
Check 1: the 3-sigma test; TunnelIntensity

A — Airehrour, Gutierrez & Ray [70]
A blackhole drops what it should forward, so ForwardingRatio and PDR collapse
Check 1: the two blackhole primaries

R — Ramírez Gómez, Vargas Montoya & León Henao [72]
Duplicate packets are the wormhole's signature
Check 1: DuplicateRate, seen by the root

K — Khan et al. [22]
A healthy ESP-MESH on ESP32 delivers above 97%
Check 1: broken baseline = excluded (0.90 bound is ours)

KW — Karlof & Wagner [81]
The attacker's position decides who the victims are
Check 2: victims vs. bystanders

H — Hu, Perrig & Johnson [46]
Wormhole = a shortcut whose copies arrive with a timing mismatch
Check 2: end A above end B; the gap is reported, not scored

Four checks, in this order (slides 14–18), then a verdict (19) and a test of the verifier (20). Verdict rules fixed before any result was read.

Testing the verifier itself

PLACEBO TEST

0 of 106

false confirmations in a placebo test (second half of the baseline relabelled as attack)

SENSITIVITY

0

verdicts changed when 7 rules were each turned about 50% down and up (14 settings, 74 runs)

STRESS TEST

An absurd setting (1,000 SDs) changed 15 verdicts, so the test can fail.

NEW SLIDE

Check 1 · Two signatures

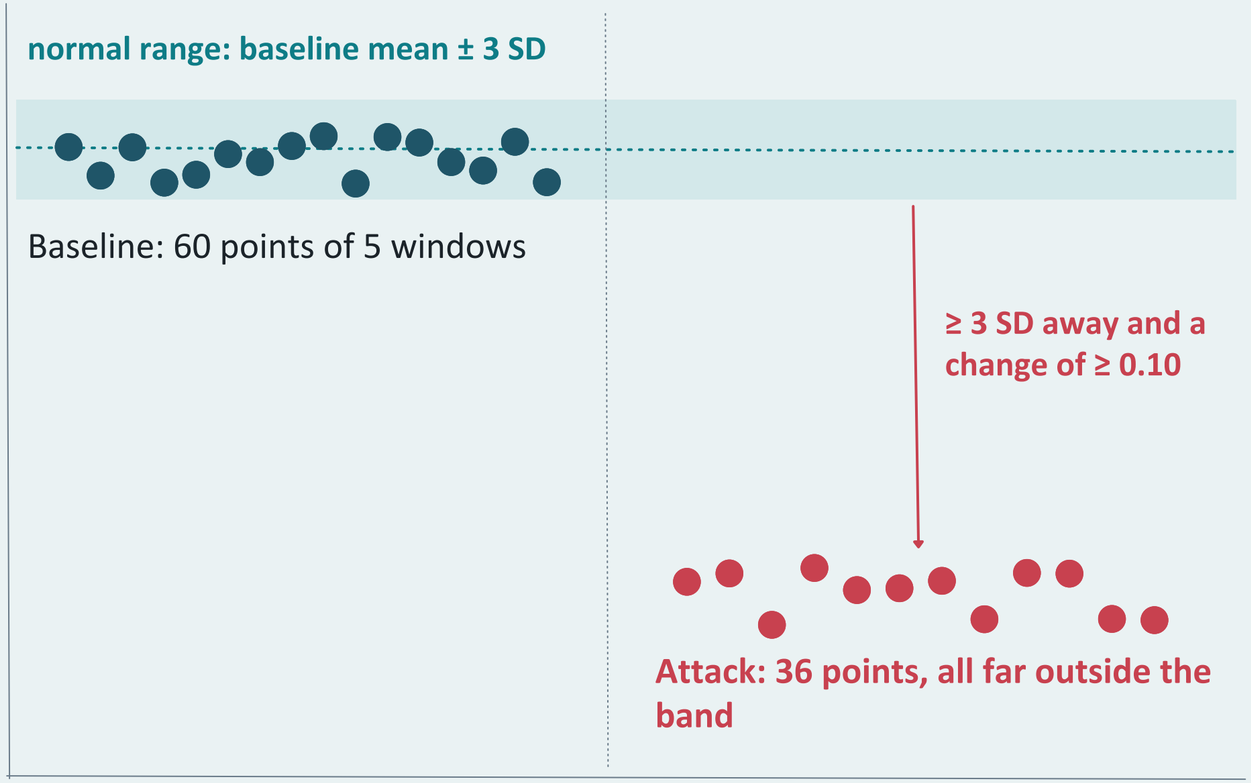
	Blackhole a relay on the path drops what it should forward	**Wormhole** a wire copies end B's probes to end A	**Benign** no attack (control)
How it works	Accepts each probe from the boards behind it, then drops it. The sender sees no error.	End B also sends each of its own probes over a UART wire to end A, which forwards it: every probe of B arrives twice.	Every board relays normally
Grounded in	Airehrour et al. [70]: forwarding and delivery collapse. Karlof & Wagner [81]: position decides the victims.	Hu, Perrig & Johnson [46]: definition. Zhukabayeva et al. [71]: out-of-band tunnel. Ramírez Gómez et al. [72]: duplicates.	Khan et al. [22]: healthy ESP-MESH > 97%
Scored (3-sigma)	ForwardingRatio ↓ at the attacker + PDR ↓ of the non-attacker boards	TunnelIntensity ↑ (the ends' own count) + DuplicateRate ↑ (seen by the root, D-21)	PDR ≥ 0.90 in both phases
Supporting	IngressEgressDelta ↑, ConsistencyScore ↑, NeighbourForwardingRatio ↓ (the neighbours' view)	TunnelBytes ↑. TunnelLatency and LatencyHopRatio are measured, not scored (no tunnel in the baseline).	Every relay forwards ≥ 90%
Who is hit	Victims behind the attacker lose every probe; bystanders are untouched. Damage = loss.	No one loses probes; only end B's probes are doubled. Damage = duplication, not loss.	No one
What we measured	36 / 36 confirmed. The attacker forwards nothing; the boards behind it lose their probes.	36 / 36 confirmed. End B's probes arrive twice; no probes are lost.	8 / 8 benign.

Same in both: no attack-caused route, parent or hop-count change, and RSSI stays flat (D-20). Both attacks are emulated at the application layer, so only traffic features move.

CHANGED

Check 1 · The 3-sigma rule

Schematic of one feature (for example ForwardingRatio) over one run, not to scale



A primary check passes when both hold

1. The attack mean is at least 3 baseline SDs from the baseline mean, in the expected direction
2. The absolute change is at least 0.10

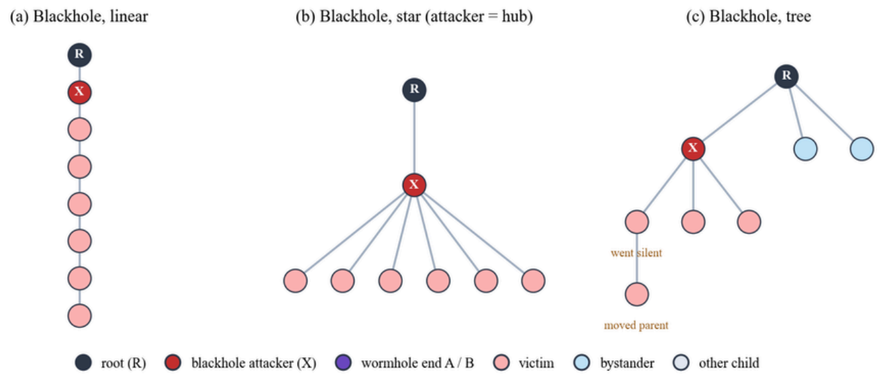
Example (illustrative numbers)

z = (0.003 − 1.000) ÷ 0.002 = −498.5

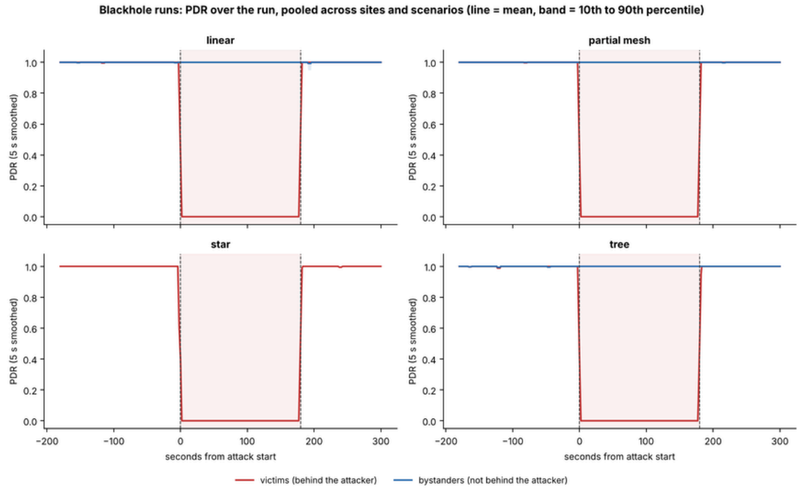
Both rules met. If the baseline never varies (SD = 0), any move in the expected direction passes.

NEW SLIDE

Check 2 · Topology



Who is a victim depends on where X sits: X = attacker, red = victims (behind X), blue = bystanders. Paper Figure 4.57 (blackhole panels)



Victims (red) lose everything in the attack; bystanders (blue) do not. Paper Figure 6.2

How the topology is checked

THE RULE

Rebuild the tree from parent addresses. Boards behind the attacker are victims; the rest are bystanders, the control group. The check reads only parent addresses, never the outcome, because a lost probe can be plain noise.

Roles first. The result never picks the victims.

THE LAYOUT CHECK

The rebuilt tree must match the topology we designed for the run: star, linear, tree or partial mesh.

THE SETUP

Blackhole: the attacker relayed, then dropped; victims stayed alive but cut off. Wormhole: one end A and one end B, A shallower. Backed by Karlof and Wagner [81] and Hu et al. [46].

NEW SLIDE

Check 3 · Dataset

1 · Count

Every board needs windows for at least 95% of every phase: baseline, attack, cooldown.

→

2 · Find the gaps

Below 95% = a gap. No file at all: the 3-sigma test decides. No attacker file = FAILED.

→

3 · Impact check

For every gap board the verifier re-runs the 3-sigma test without it, automatically. Does the attack still confirm?

Then the grade: decided by the re-run and by the board's role

STRONG / GOOD	MINOR GAPS	MAJOR GAPS	FAILED
No gap boards. All board data is present. The run is validated.	Without the gap boards the attack still confirms, and at most half the boards have gaps. Kept and footnoted. Recapture advised.	Without the gap boards the attack no longer confirms, or over half the boards have gaps. Recapture.	The attack cannot be confirmed: no signature, too little data to test, or the attacker is missing. Redo.

Disconnected board
Layer -1, no parent: its PDR is blank for those seconds, never 0. Under 95% of windows: a gap board, so the re-run applies.

No file (board missing)
Nothing to re-run: no data, nothing to remove. The 3-sigma test already ran without it. Attack still confirmed = minor gap, even if a victim or bystander is missing. Test fails = FAILED. Attacker file missing = FAILED, even at Check 1.

Our own rule, refined during capture (D-32). Missing data is never silently passed.

NEW SLIDE

Check 4 · Scenario

Did the traffic leave the trace that its scenario should leave?

STATIONARY

Each child probes about once per second.

Check: the children's median rate is at most 1.5 probes per second.

HIGHLOAD

A probe every 250 ms, about 4 per second instead of 1.

Check: the children's median baseline rate is at least 3 probes per second.

BURST

One child fires 300 probes in about 3 s, 60 s into the attack.

Check: at least 240 extra probes from one child within 10 s.

TRACE FOUND

The scenario is APPLIED. The run counts for the scenario it is filed under.

TRACE MISSING

NOT APPLIED: a minor gap, footnoted (D-34). The attack result stands; the scenario needs a recapture.

NEW SLIDE

The verdict

1 Signature | 2 Topology | 3 Dataset | 4 Scenario → VERDICT

PASS

The signature is confirmed by the 3-sigma test on the primaries, and topology, dataset and scenario hold.

Graded STRONG, GOOD or MINOR GAPS (kept, footnoted).

FAIL

No signature, a board changed class, the attacker's data is missing, or the setup did not run as designed.

The run is redone.

INCONCLUSIVE

The evidence cannot decide. A broken baseline (mean below 0.90 or SD above 0.15 of the mean) excludes that primary: excluded, not failed.

Healthy ESP-MESH delivers above 97% (Khan [22]); we set our 0.90 line lower on purpose.

WHAT THE SIGNATURE SCORES

Blackhole: the attacker's ForwardingRatio and the PDR of the non-attacker boards.
Wormhole: TunnelIntensity (the ends' own count) and DuplicateRate (the root's own view, D-21).

NEW SLIDE

Testing the verifier

A validator is only believable if it can say no.

PLACEBO TEST

0 of 106

false confirmations when the second half of the baseline was relabelled as attack.

SENSITIVITY

0

verdicts changed when we turned each of our 7 rules about 50% down and up (14 settings, 74 runs). Details: backup slide 21.

STRESS TEST

15

verdicts changed with an absurd setting (1,000 SDs): the test can fail, so a pass means something.

Rules were fixed before any result was read. Missing data never becomes a pass or a fail.

Backup: the 17 sensitivity settings

7 rules, each turned about 50% down and up = 14 settings, plus the unchanged verifier and 2 stress controls = 17 settings. SD = standard deviation (how much the normal stretch wobbles). PDR = share of probes that reached the root (0 to 1). ForwardingRatio = forwarded ÷ received (0 to 1).

The rule (plain words)	Feature it checks	Limit used (with unit)	Tested at
Abnormal = this far from normal	All 4 scored features	**3 standard deviations**	2.5 and 3.5 SDs
Smallest change that counts	All 4 scored features	**0.10 (on a 0 to 1 scale)**	0.05 and 0.20
Victims must lose at least half	Victims' PDR	**PDR below 0.50 (50%)**	0.25 and 0.75
Bystanders may lose at most 30%	Bystanders' PDR	**PDR falls by up to 0.30**	0.15 and 0.45
Attacker must pass on under half	Attacker's ForwardingRatio	**Ratio below 0.50 (50%)**	0.25 and 0.75
Time allowed in the wrong group	Victim / bystander role	**3 seconds**	2 and 5 seconds
Normal stretch must be this good	PDR + ForwardingRatio	**At least 0.90 (90%)**	0.80 and 0.95

Result: 0 verdicts changed in the 14 rule settings (74 runs). Absurd control (1,000 SDs): 15 changed.

Chapter 5: 37 deviations, three reasons

37

documented deviations, D-1 to D-37

17 modify what the proposal specified; 20 add a rule the proposal did not have. Each lists what was planned, what changed, why, and the cost (Appendix C).

1

The platform could not measure it

No MAC retry count reaches the app (D-15); probes get no reply, so no round-trip time (D-2, D-3)

2

The plan would leak the label or fail to make the attack

Relay counters existed only on the attacker (D-12); ends at the same depth skip no hop (D-17, D-28, D-30)

3

A run needed rules to be one valid experiment

Pre-run gates and START (D-22); parent changes in the attack (D-23); missing boards by role (D-32)

Planned vs. actual: the changes that matter

The proposal planned	What we did	Why
Attacks on routing (false route, false shortcut)	App-layer emulation: a placed relay drops; a wire copies probes	The API reads routes but cannot write them (D-20)
The mesh stack relays traffic out of sight	Every node relays probes itself and counts them	Only the attacker had relay counters, which leaked the label (D-12)
MAC retry count from Wi-Fi statistics	App counters kept as a column; real retries from a sniffer (7 runs)	No retry number reaches the app (D-15)
5 s windows of 5 samples	1 s windows of 10 samples at 10 Hz	One lost sample cost 20% of a window (D-1, D-9)
Flat star; fixed wormhole ends	Hub = attacker or end A; ends chosen by depth, B deeper	Root cannot attack; equal depth skips no hop (D-16, D-17, D-28, D-30)
No benign control runs	8 benign runs across sites and scenarios	A control for false alarms (D-18, D-35)

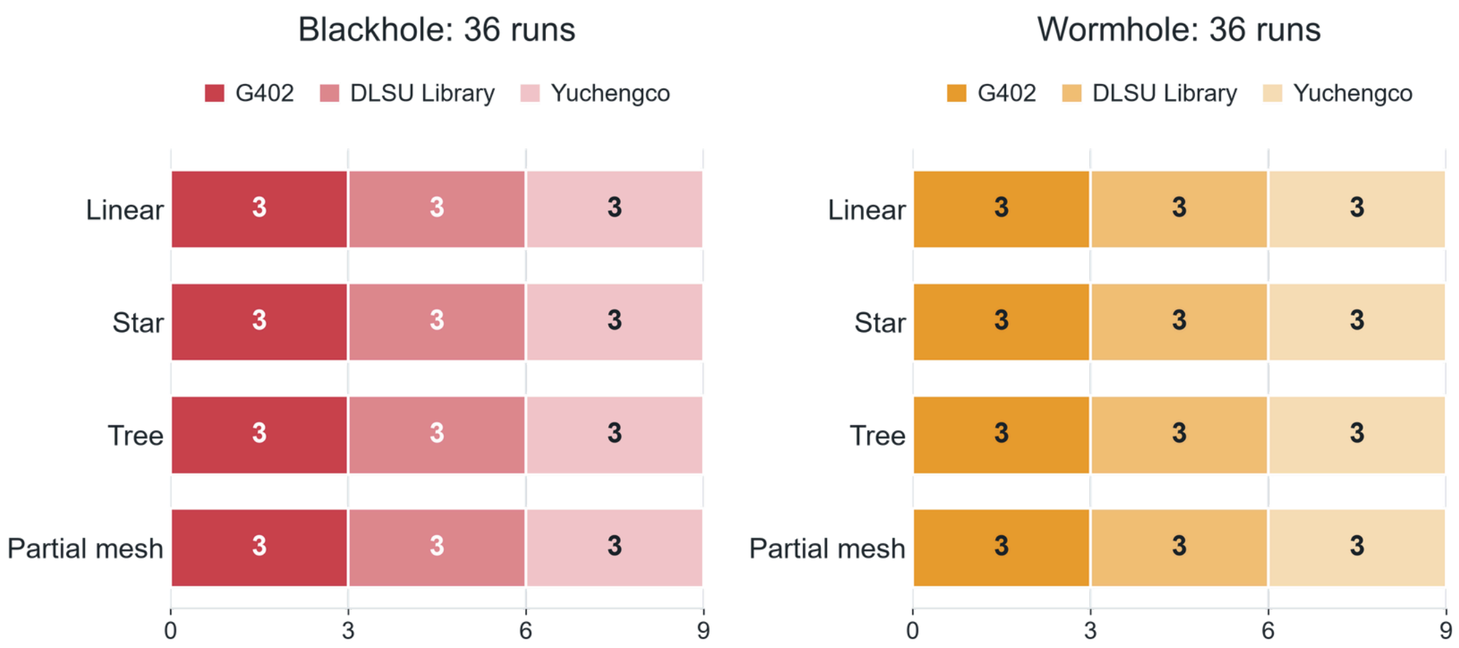
Limits how Chapter 6 is read (5.7): D-20 emulation, D-15 no MAC retry count, D-18/D-35 only 8 benign runs, D-37 tree fan-out differs per run.

6

Chapter 6: Results and Discussion

The dataset, attack validation, EDA, clustering and what it all means

The dataset: 80 of 80 runs captured



80 / 80 runs: 36 blackhole, 36 wormhole, 8 benign

369,678 windows analysed (of 456,195)

0.30% windows discarded (1,372)

637 log files, none empty

Benign (no attack): 8 runs. Linear 3, star 3, tree 1, partial mesh 1. Each one shares its topology, site and scenario with an attack run.

Analysed: 266,491 normal + 103,187 attack windows. Blackhole boards: 123 victims, 92 bystanders (the star has none).

Question: How many runs did we capture for each attack?

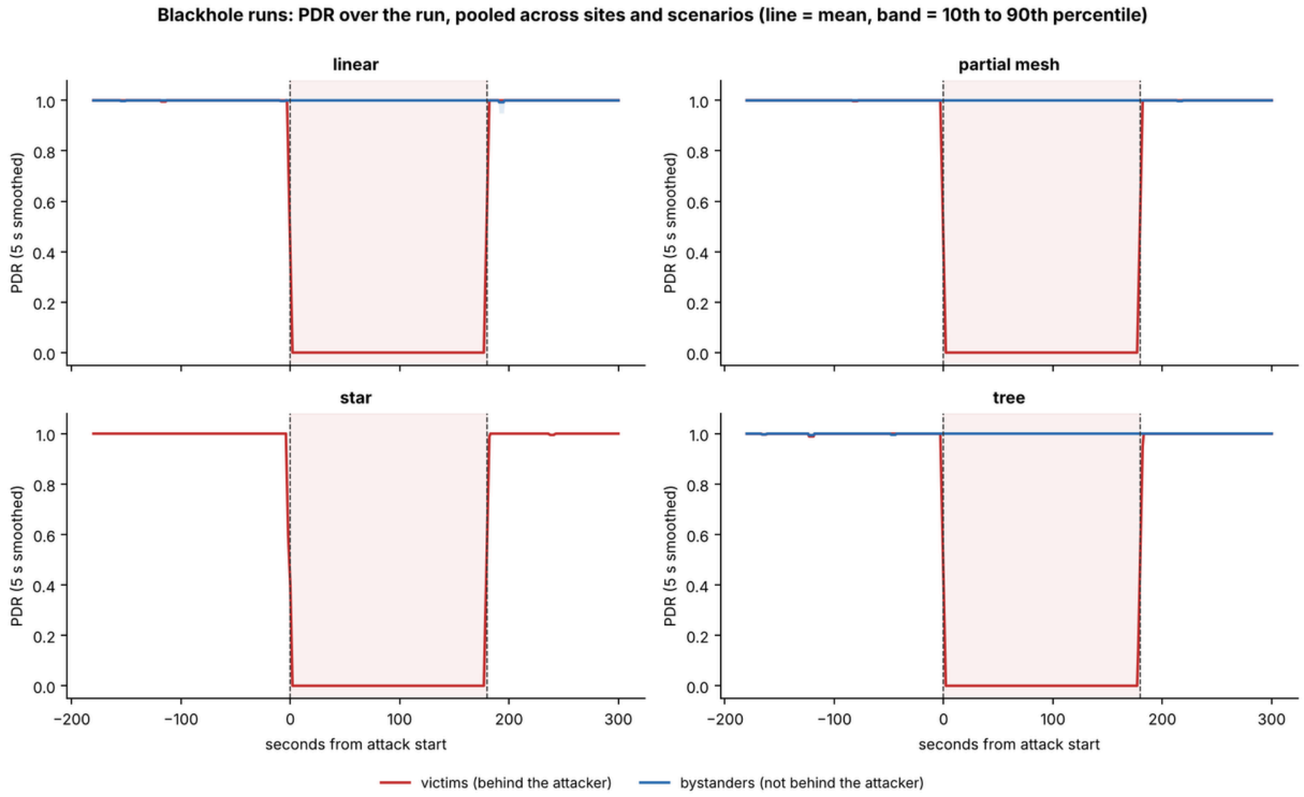
Left = blackhole, right = wormhole. One bar per topology, split by site (shades). Each part = 3 runs, one per traffic scenario (stationary, burst, highload).

6A

Blackhole results

Did the attack work, what it changed, and whether clustering finds it

Blackhole: confirmed in all 36 runs



36 / 36

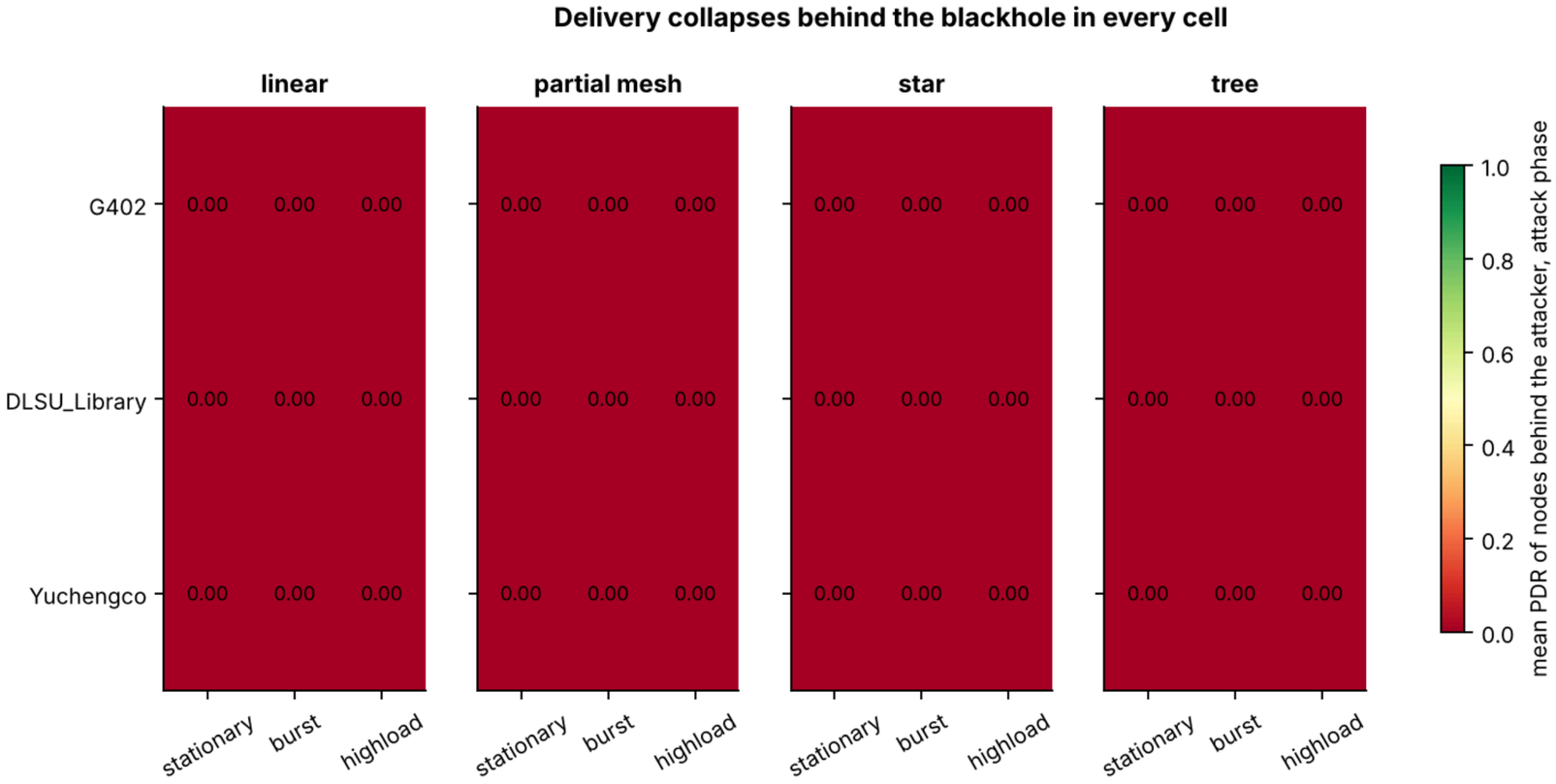
blackhole runs confirmed (4 strong, 27 good, 5 minor gaps)

- Attacker ForwardingRatio: 1.000–1.004 → 0.000–0.007
- Victim PDR: 0.998 → 0.000; every one of the 123 victims averaged exactly 0
- 121 of 123 victims cut off in the first second; recovery median 2 s
- Bystanders: 0.996 → 1.000

Question: Did the victims lose delivery during the blackhole, and only then?

Time runs left to right (0 = attack starts, 180 = attack ends, shaded). Height = PDR, the share of probes that reached the root (1 = all). Red = victims, blue = bystanders.

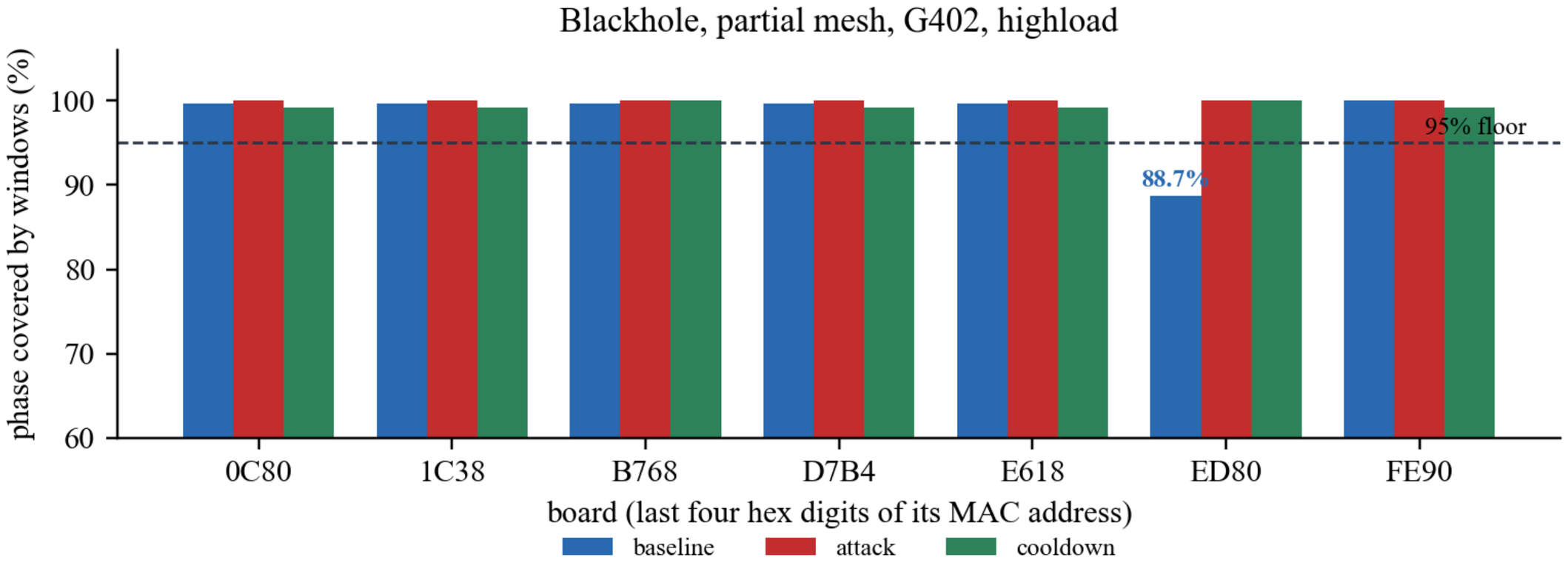
Figure 6.1: delivery collapses in every blackhole run



Question: Did the blackhole cut off the victims in every single run?

Each tile = one run (panel = topology, row = site, column = scenario). Colour and number = the victims' average PDR during the attack: green = 1 (all delivered), dark red = 0 (nothing).

Figure 4.58: dataset check, one run



Question: Did every board record enough of each phase?

One group of bars per board (last four hex digits of its MAC). Height = % of the phase it recorded. Blue = baseline, red = attack, green = cooldown. Dashed line = the 95% minimum.

Obersevation: Every bar clears 95% except board ED80's baseline (88.7%). That is a minor gap, so the run is kept with a footnote.

Blackhole predictions: delivery yes, retransmissions no

Prediction (Table 3.4)	Observed?	Measured
Attacker: ForwardingRatio < 1	**Yes**	1.000–1.004 → 0.000–0.007; passed 36 of 36 runs
Attacker: Ingress/Egress Delta > 0	**Yes**	Passed 31 of 36; the other 5 rose but stayed under 3 SD
Victims: PDR falls	**Yes**	0.998 → 0.000; all 123 victims averaged exactly 0
Victims retransmit more (MAC)	**No**	App counter 0.0000 → 0.0004; sniffer (7 runs): 3 up, 4 down
Attacker: RSSI stays stable (PHY) †	**Yes**	−67.1 → −67.8 dBm; within 1 dB in 27 of 36 runs

† Not reported in the paper: computed from the same feature tables (attacker windows, baseline vs. attack). Table 3.4 rows 1 to 4: Tables 6.9 and 6.10, Section 6.2.4.

The attacker's counters and the victims' delivery moved; the radio never saw a failure, because the attacker still acknowledges every frame.

Clustering: how the data were asked to sort themselves

5 Algorithms

- K-Means
- Ward (hierarchical)
- GMM (Gaussian mixture)
- DBSCAN (eps at the k-distance knee)
- Spectral

3 Feature sets

- FULL: 8 features allowed by the leakage guard
- REDUCED: PDR, RSSI_mean, RSSI_var, RSSI_Hop_Diff
- Without PDR: the 3 RSSI features, as a control

4 Scores

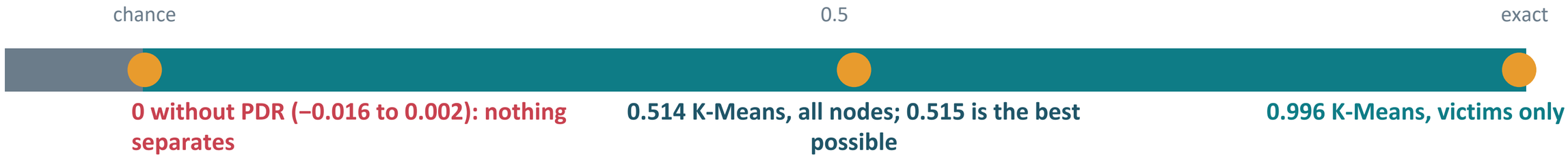
- Silhouette, Davies–Bouldin: cluster shape, no labels
- Purity, ARI: agreement with the label, computed only afterwards
- ARI: 1 = exact, 0 = chance

6,000 sampled windows per seed (42, 7, 123), standardised features, k = 2 to 8. The algorithm never sees the label.

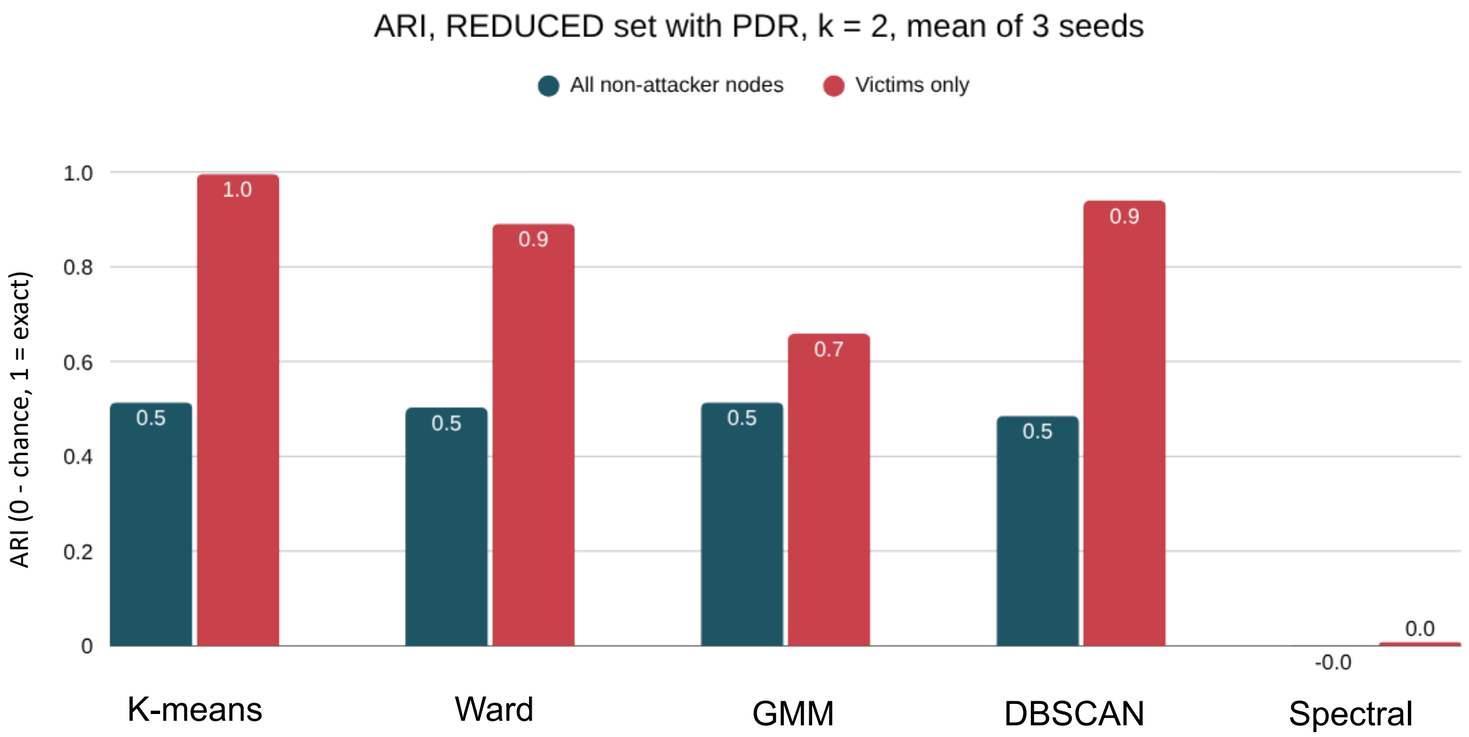
How to read the clustering scores

Score	Question it answers	Uses the label?	How to read it
Silhouette	How close is a window to its own cluster compared with the nearest other one?	No	−1 to 1, higher = tighter. Describes shape, not the attack
Davies–Bouldin	How compact and how far apart are the clusters?	No	0 and up, lower = better. Same caveat as the silhouette
Purity	What share of windows carries the majority label of their cluster?	Yes, afterwards	Compare with the majority rate: 0.70 = no separation
ARI	Do the clusters agree with the label, corrected for chance?	Yes, afterwards	1 = exact, 0 = chance. Here the best possible is 0.515

The ARI scale, with the blackhole results on it



Blackhole clustering: PDR carries all the separation



0.514

K-Means ARI on all non-attacker nodes; best possible is 0.515

≈ 0

every algorithm, every k, without PDR (−0.016 to 0.002)

Remove one feature and the separation disappears: the blackhole leaves no mark on signal strength or topology.

How to read it

Question: How well does each clustering method recover the blackhole label?
How to read: One pair of bars per method. Height = ARI: 0 = no better than chance, 1 = perfect match. Dark blue = all non-attacker nodes, red = victims only (REDUCED set, k = 2, mean of 3 seeds).
What you see: On victims, K-Means matches almost perfectly (0.996). On all nodes every method stops near 0.5, the best possible here, because bystanders carry the attack label but behave normally.

Why 0.514 is the best possible score

Attack-labelled windows of non-attacker nodes (38,232)

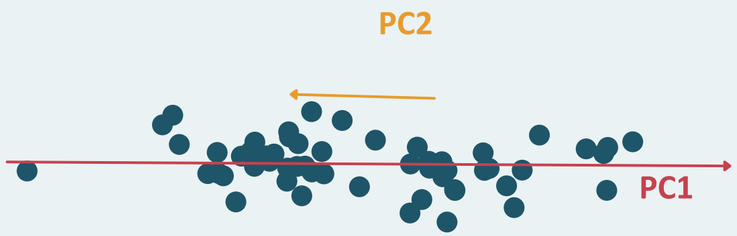


1. The phase label marks every board of a blackhole run as "attack", bystanders included
2. A bystander's PDR stays at 1.000, so no feature can show an attack that never reached it
3. Best possible split: victim attack windows vs. everything else. Its ARI is 0.515, purity 0.870

0.514	K-Means, all non-attacker nodes
0.515	best possible on these labels
0.996	K-Means, victims only

PCA and t-SNE: how to read the pictures

PCA



- Rotates the data so axis 1 holds the most variance, axis 2 the next
- Axes are in standardised feature units; the percentage on an axis is the variance it holds
- A few extreme windows can stretch it into a thin column (Figure 6.22)

t-SNE

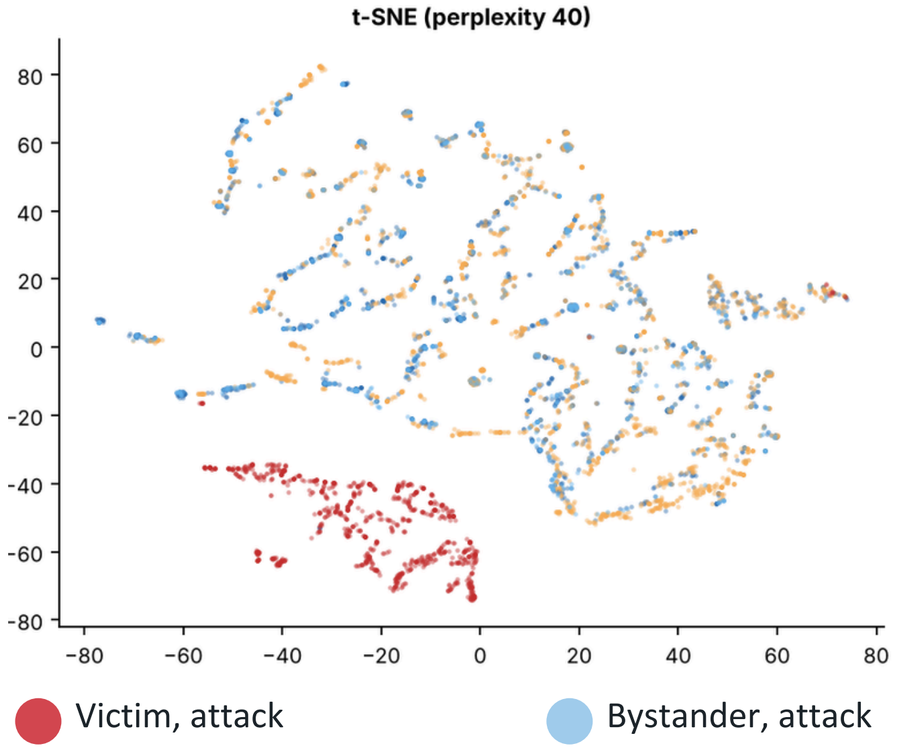


- Places windows with similar features near each other
- Axes have no units; only nearness means anything
- Never read the distance between far islands

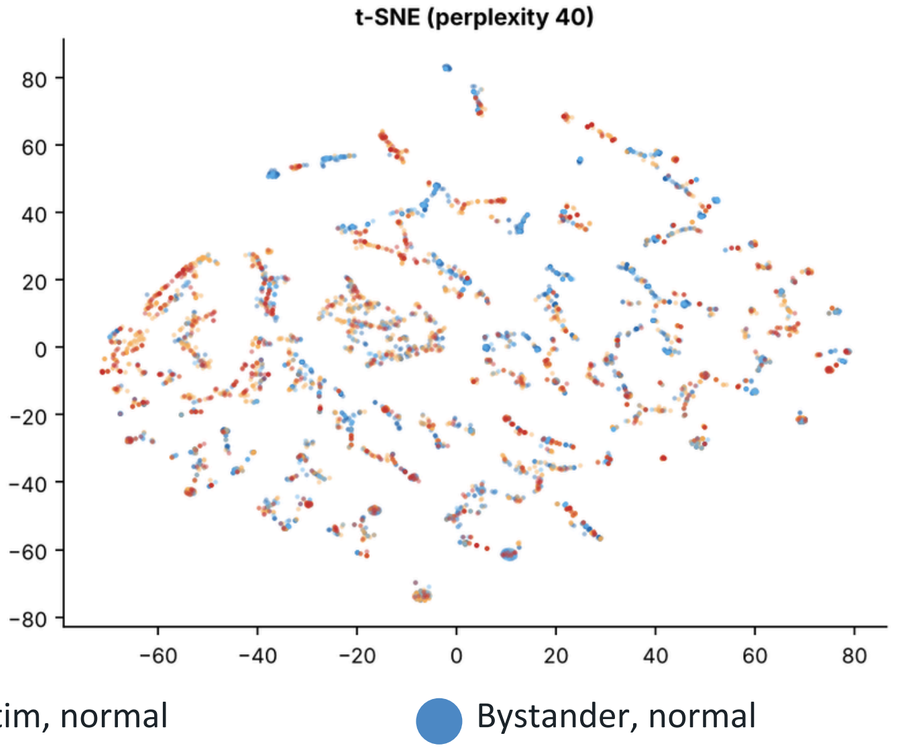
A group in a picture is not a cluster; a cluster is what an algorithm returns. We use these pictures to describe the data, and the scores decide whether a separation exists.

The same windows, with and without PDR

With PDR (FULL set, 8 features)



PDR removed (3 RSSI features)



Victim, attack

Bystander, attack

Victim, normal

Bystander, normal

Question: Do the attack seconds form their own group, with and without PDR?

Each dot = one second of one board (6,000 sampled). Dots close together have similar values; the axes have no units. Red = victim attack, light blue = bystander attack, orange = victim normal, dark blue = bystander normal.

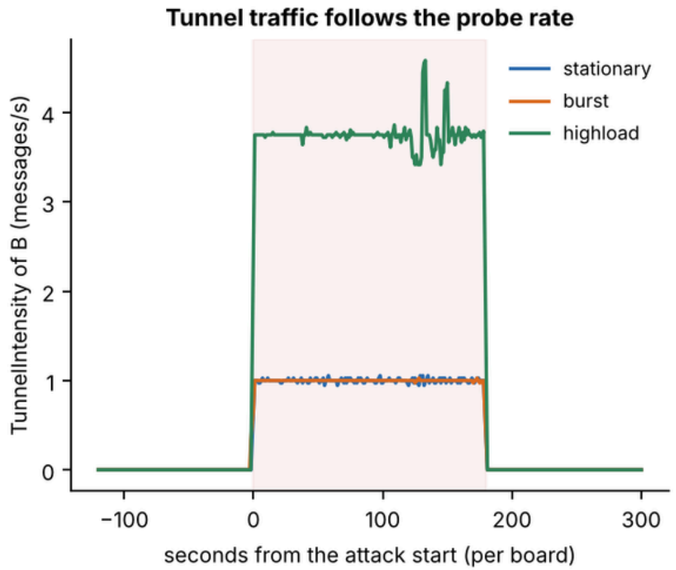
With PDR (left) the red dots form their own island. Without PDR (right) all four colors are mixed: there is no attack group to find.

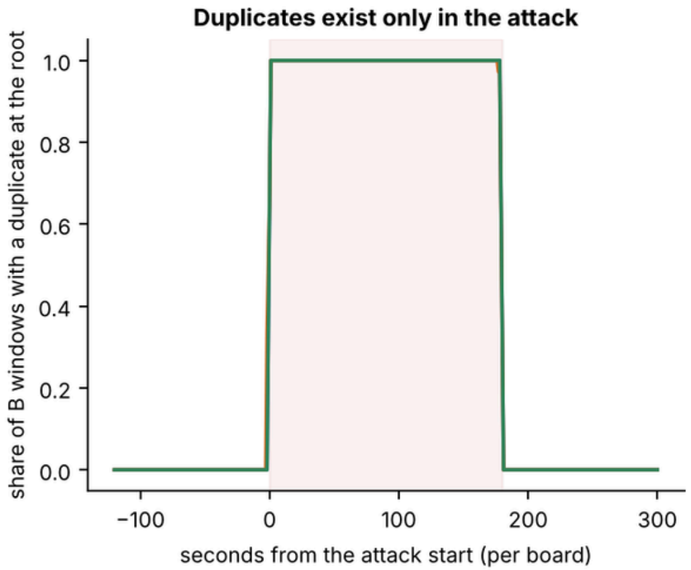
6B

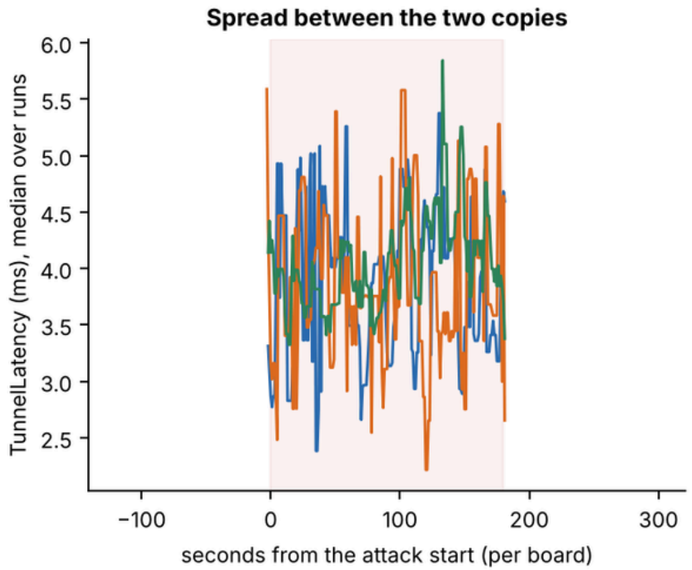
Wormhole results

Did the tunnel work, what it changed, and what clustering finds

Wormhole: confirmed in all 36 runs







36 / 36

wormhole runs confirmed (21 strong, 15 good)

99.86%

of end B's probes reached the root twice (12,448 of 12,465)

0

duplicates in baseline or cooldown, and none from other boards

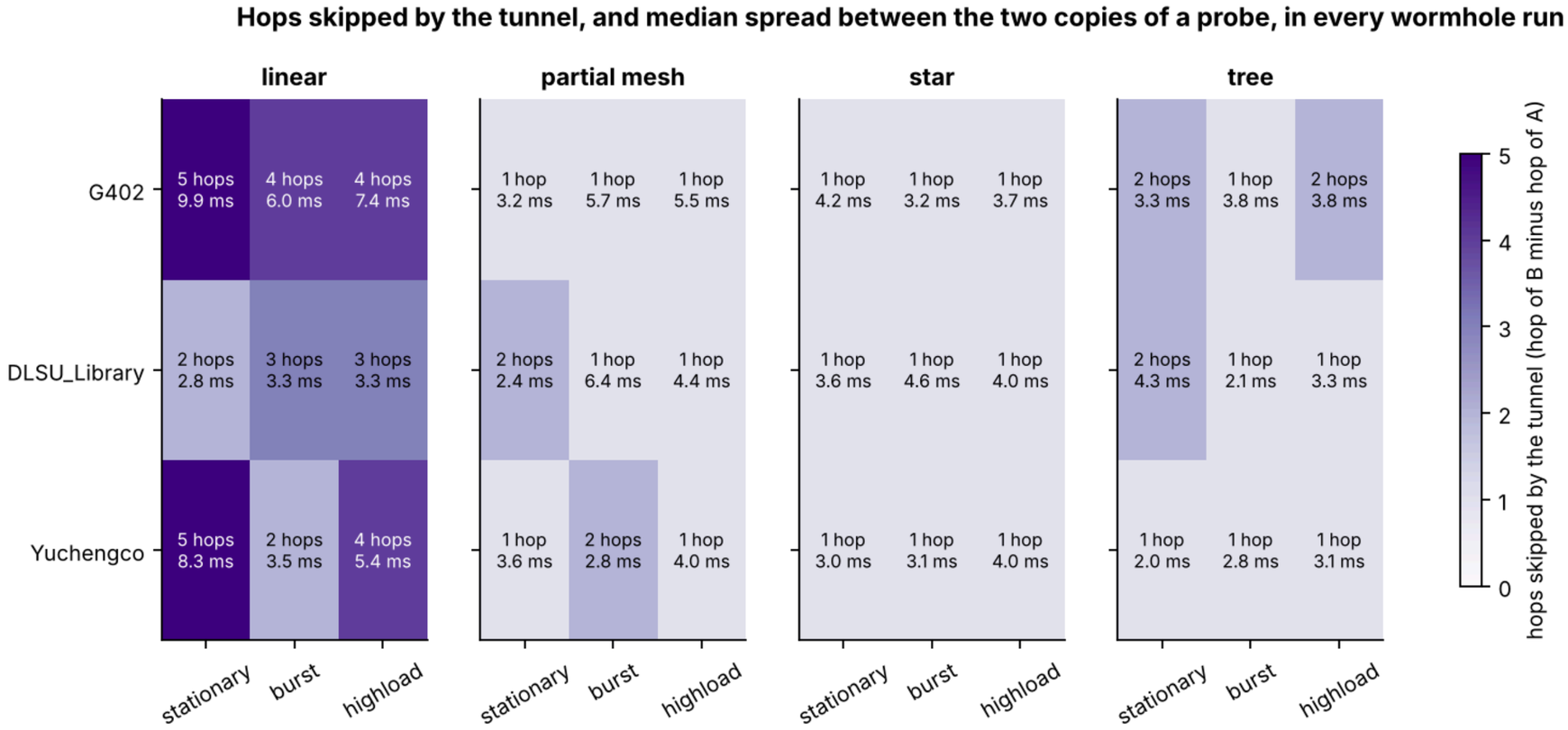
3.9 ms

median gap between the two copies of a probe

How to read it

Question: Did the tunnel work, and only during the attack?
How to read: Shaded = the 180 s attack. Left: end B's tunnel messages per second. Middle: share of B's seconds with a duplicate at the root (0 to 1). Right: gap between the two copies (ms). Colour = scenario.
What you see: Left and middle are 0 before and after the attack and jump up only inside it (highload about 4 messages/s). The gap exists only during the attack and is a few ms.

Figure 6.4: hops skipped by the tunnel, every wormhole run

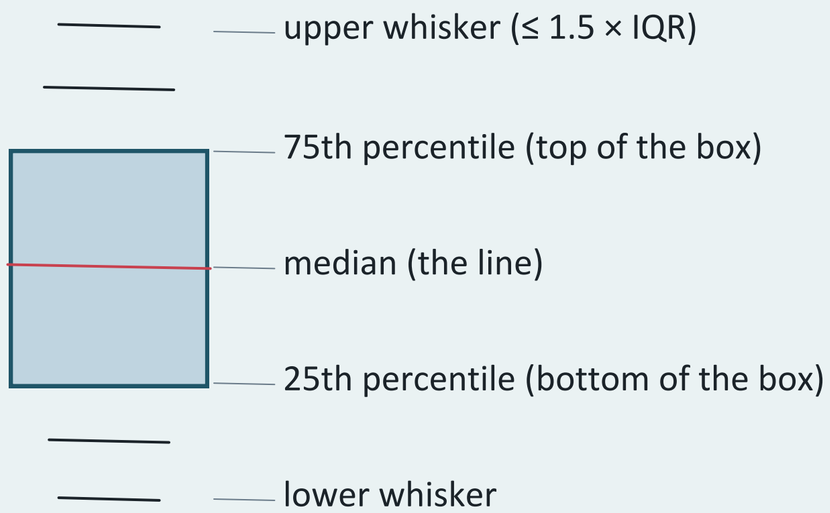


How to read it

Question: How many hops did the tunnel skip in each wormhole run, and how far apart did the two copies arrive?
How to read: Each tile = one run (panel = topology, row = site, column = scenario). Darker purple = more hops skipped (1 to 5). The second number = the typical gap between the two copies (ms).
What you see: Only linear skips many hops (2 to 5); the star always skips 1. Gaps stay between 2 and 10 ms. The 5-hop linear tiles are the highest, but across all runs there is no clear trend (Figure 6.6).

How to read box plots and Cohen's d

A box plot (Figures 6.9, 6.12, 6.15)



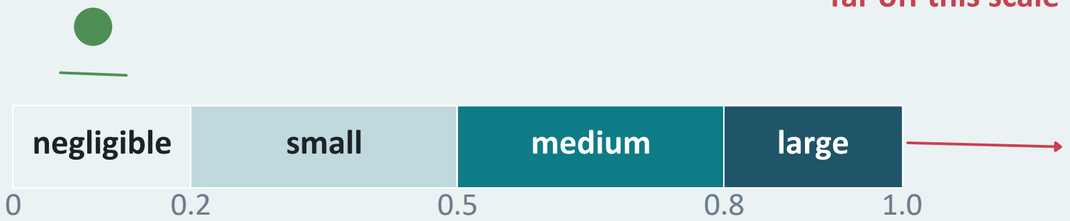
Box = IQR, the middle half of the values. Blue = baseline, red = attack. Boxes that overlap mean the feature did not change.

Cohen's d: how big is the shift?

d = (attack mean − baseline mean) ÷ pooled SD

most features: |d| < 0.1

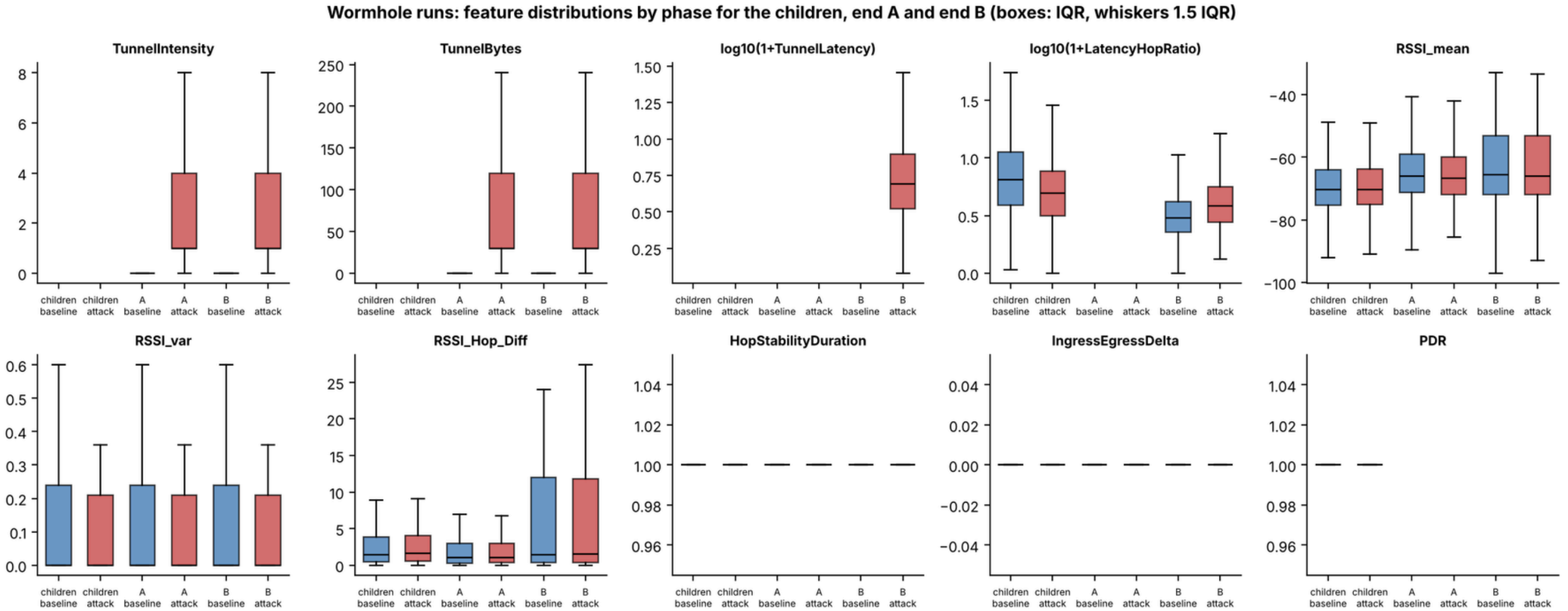
blackhole victim PDR: |d| = 29.71, far off this scale



- Negative = lower during the attack
- A size, not a significance test: the windows of one board follow each other, so they are not independent
- Wormhole ends: TunnelIntensity d = 1.83; wormhole non-attacker nodes: largest |d| = 0.06

Worked check: d = −29.71 with means 0.998 → 0.000 implies a pooled SD of about 0.034 (0.998 ÷ 29.71). The mean moved almost 30 typical spreads.

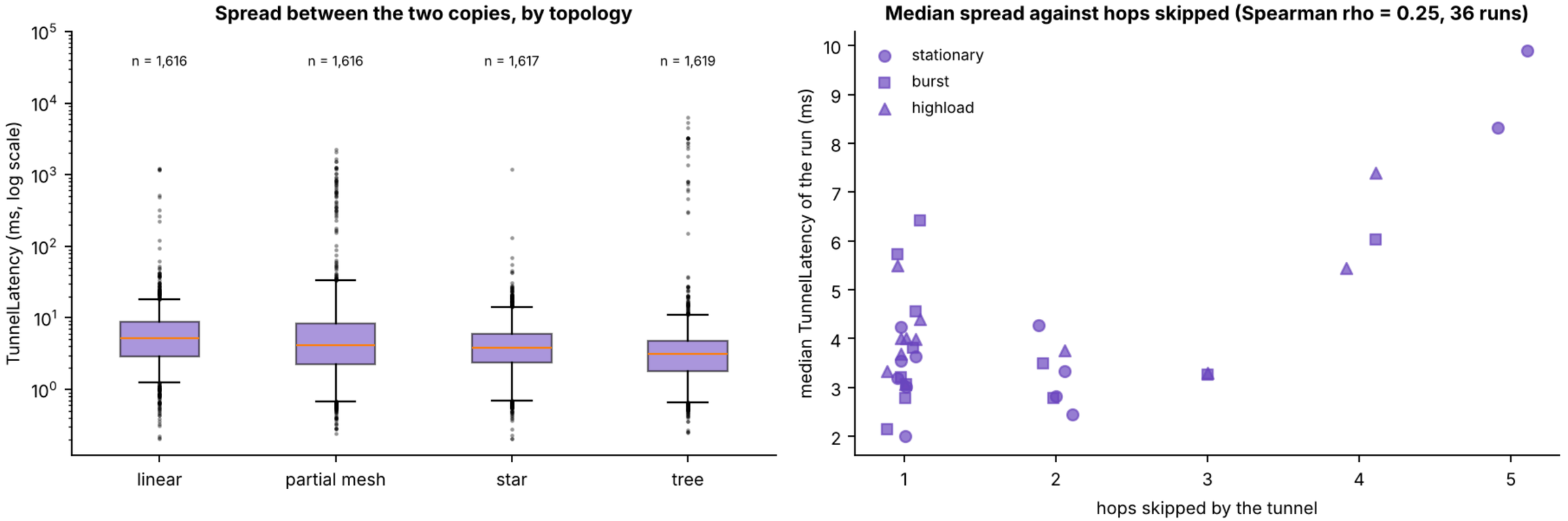
Figure 6.12: wormhole feature distributions by phase



How to read it

Question: Which features change in the wormhole, and on which boards?
How to read: One panel per feature; boxes for the other children, end A and end B, each baseline (blue) then attack (red). The tunnel features exist only on the two ends.
What you see: Only the tunnel panels change, and only on A and B during the attack. The signal-strength boxes are the same before and during the attack.

Figure 6.6: spread between the two copies of a probe



How to read it

Question: How far apart do the two copies of a probe arrive, and does that grow with the hops skipped?
How to read: Left: one box per topology; the y-axis is logarithmic (each step = ×10). Box = middle half, whiskers = 5th to 95th percentile, dots = rare extremes. Right: one dot per run, hops skipped vs. typical gap; shape = scenario.
What you see: The typical gap is a few ms in every topology, with rare long delays. The 4- and 5-hop runs sit a bit higher, but the link is weak and not significant (Spearman 0.25, p = 0.14).

Wormhole predictions: traffic yes, radio and routing no

Prediction (Table 3.5)	Observed?	Measured
Hop counts change	**No**	0 windows with a hop change in any phase
Impossible parent links	**No**	0 windows with a parent change
RSSI does not fit the depth	**No**	RSSI_Hop_Diff A 2.79 → 2.76 dB; B 5.41 → 5.68 dB
RSSI variance rises	**No**	Cohen's d −0.03 (A), −0.07 (B)
Duplicate arrivals at the root	**Yes**	6,468 of 6,469 attack windows of end B
Latency mismatch between copies	**Yes**	Median spread 3.9 ms

What belongs to the traffic the tunnel creates was observed; what belongs to routing and radio was not.

Wormhole and pooled clustering

Wormhole: cross-layer features find nothing

- Non-attacker boards: ARI −0.012 to 0.003, every algorithm and k
- Tunnel ends with the RSSI features: −0.012 to 0.009
- Diagnostic only, TunnelIntensity added: DBSCAN reaches 0.827 on end B
- The tunnel signal exists, but in the attackers' own counters, not in the cross-layer features

Pooled: 80 runs, three labels

0.333 K-Means ARI; best possible 0.334

As a detector (one model applied to all windows):

Group	In the "victim" cluster
Victim attack windows	21,733 of 21,733
Benign windows	0 of 22,481
Normal, blackhole runs	183 of 89,140
Wormhole attack windows	3 of 32,194

6C

Benign runs and both attacks

The control runs, the single-feature audit, the side-by-side comparison and what it all means

Benign runs: no attack signature

8 / 8

benign runs confirmed benign, all strong

0 of 21

evaluable signature checks passed

24 of 24

cooldown comparisons within 0.5 baseline SD

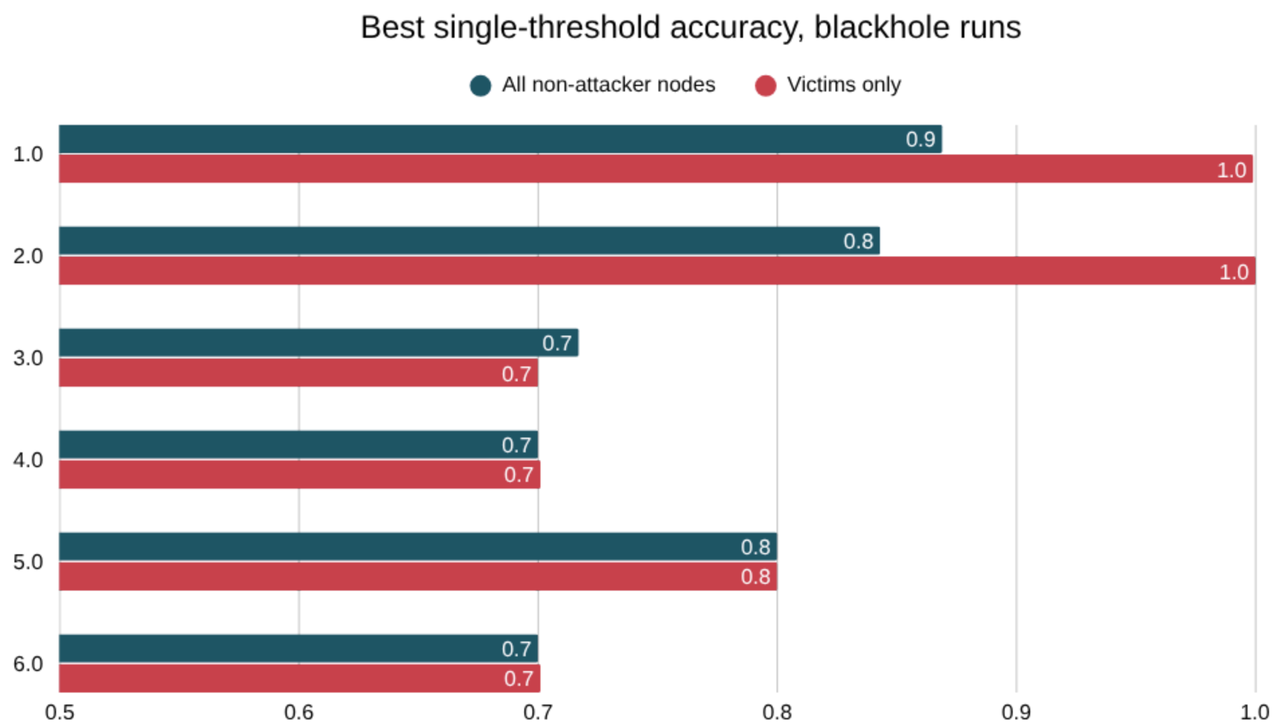
? How a benign run is tested

- Same tests, pointed the other way: cooldown against baseline
- No signature may pass; delivery must stay healthy (PDR ≥ 0.90, every relay forwards ≥ 90%)
- Mean PDR per run: 0.9994 to 1.0000

! What 8 runs can show

- Enough as a control: none triggered the verifier
- Benign runs were done just for suplementation to the normal activity set on the runs

Can one feature decide it? The single-feature audit

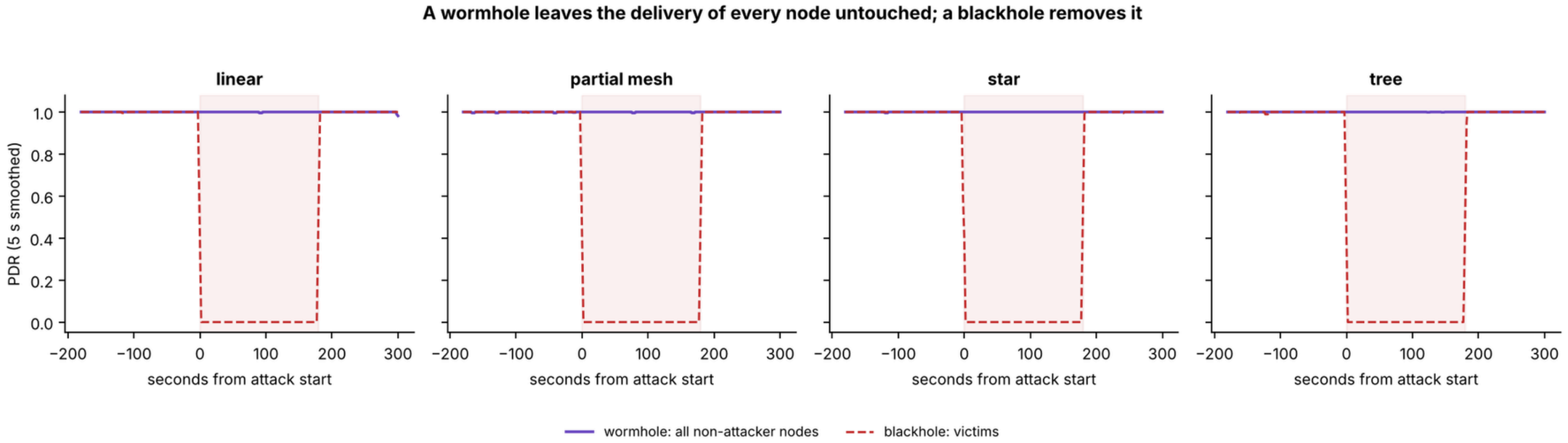


- Majority rate ≈ 70%: 420 normal s of 600 s
- PDR separates victims at 99.9%: a full delivery collapse is separable by construction
- Only 86.9% on all nodes: bystanders carry the attack label but behave normally
- LatencyHopRatio is missing when a probe is dropped, which gives the label away, so it is not clustered
- Wormhole: on non-attacker nodes no feature beats 70% by more than 0.01 points

How to read it

Question: Can one feature with one cut-off tell attack seconds from normal ones?
How to read: Each bar = the best accuracy of a single cut-off on that feature (axis starts at 50%). Dark blue = all non-attacker nodes, red = victims only. † = judged only where the feature has a value.
What you see: 70% means "no information": always guessing "normal" already scores 70%. PDR stands out (99.9% on victims). LatencyHopRatio wins only through its blanks, and the run clock (80%) is the fixed schedule, not a feature.

EDA: only the delivery of the victims moves



Blackhole: victim PDR 0.998 → 0.000, Cohen's d = −29.71. Every other feature of victims and bystanders: |d| < 0.1.

Wormhole: non-attacker PDR 0.9999 → 0.9999; largest |d| = 0.06. On the ends only the tunnel features change (0 → 1.92 msg/s).

How to read it

Question: Does the wormhole hurt delivery the way the blackhole does?
How to read: Time runs left to right (shaded = attack). Height = PDR, the share of probes that reached the root. Solid purple = wormhole runs, non-attacker boards. Dashed red = blackhole victims.
What you see: Purple stays at 1 for the whole run; red drops to 0 for the whole attack. The wormhole does not stop delivery, the blackhole does.

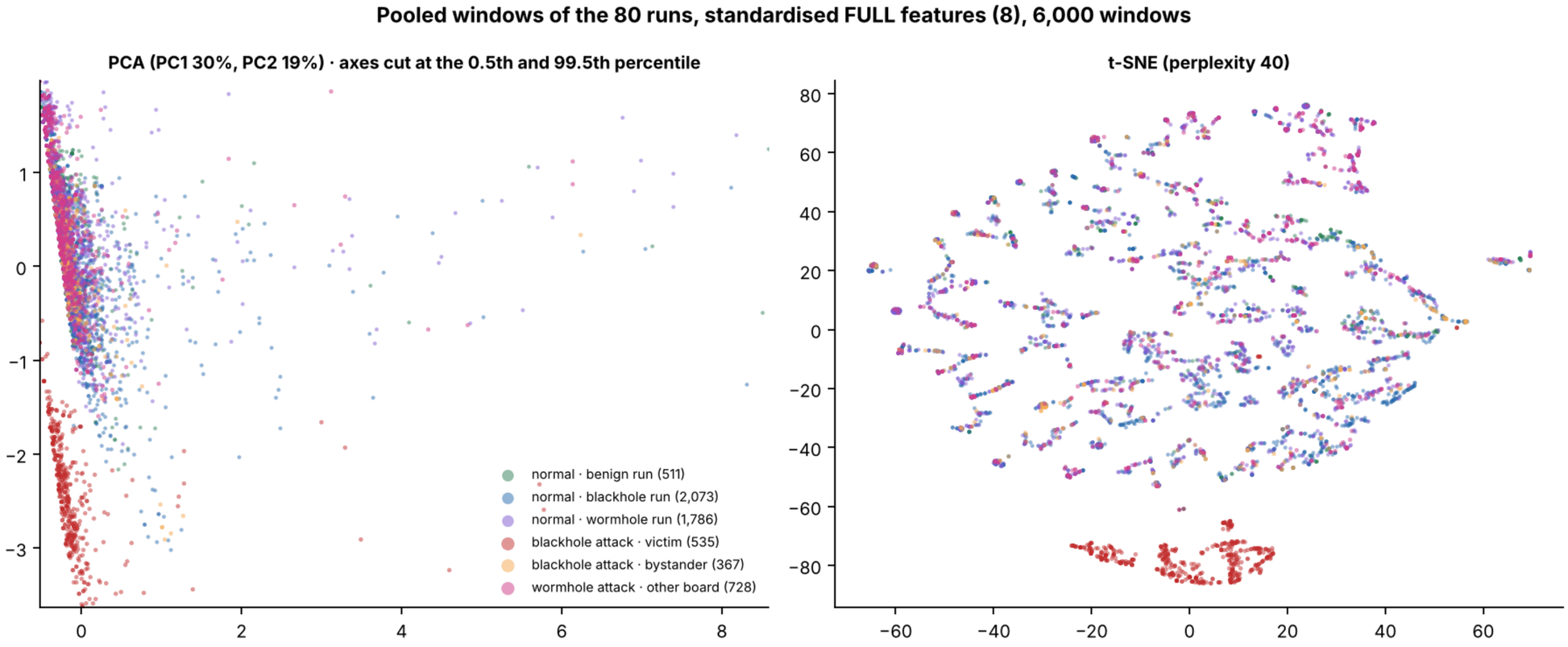
Discussion: the attacks live in the data plane

Run type	Direct record	Cross-layer features that changed	Clustering
Blackhole	Attacker FR → 0.000–0.007; victim PDR 0.998 → 0.000	Victim PDR only (d = −29.7); all others |d| < 0.1	0.514 (ceiling 0.515); ≈ 0 without PDR
Wormhole	B's probes reach the root twice (99.86%)	None (|d| ≤ 0.07; hop and parent unchanged)	≈ 0 without a tunnel feature
Benign	0 of 21 signature checks passed	Cooldown within 0.5 SD in 24 of 24	0 of 22,481 in the victim cluster

Main reading: Chapter 4 expected cross-layer evidence. The evidence sits in delivery (blackhole) and duplication (wormhole); PHY, MAC and network features stay where they were.

Honest qualifier: PDR is also the verifier's check, so the clustering finds the effect the verifier measured. It is not independent evidence that cross-layer features detect the blackhole.

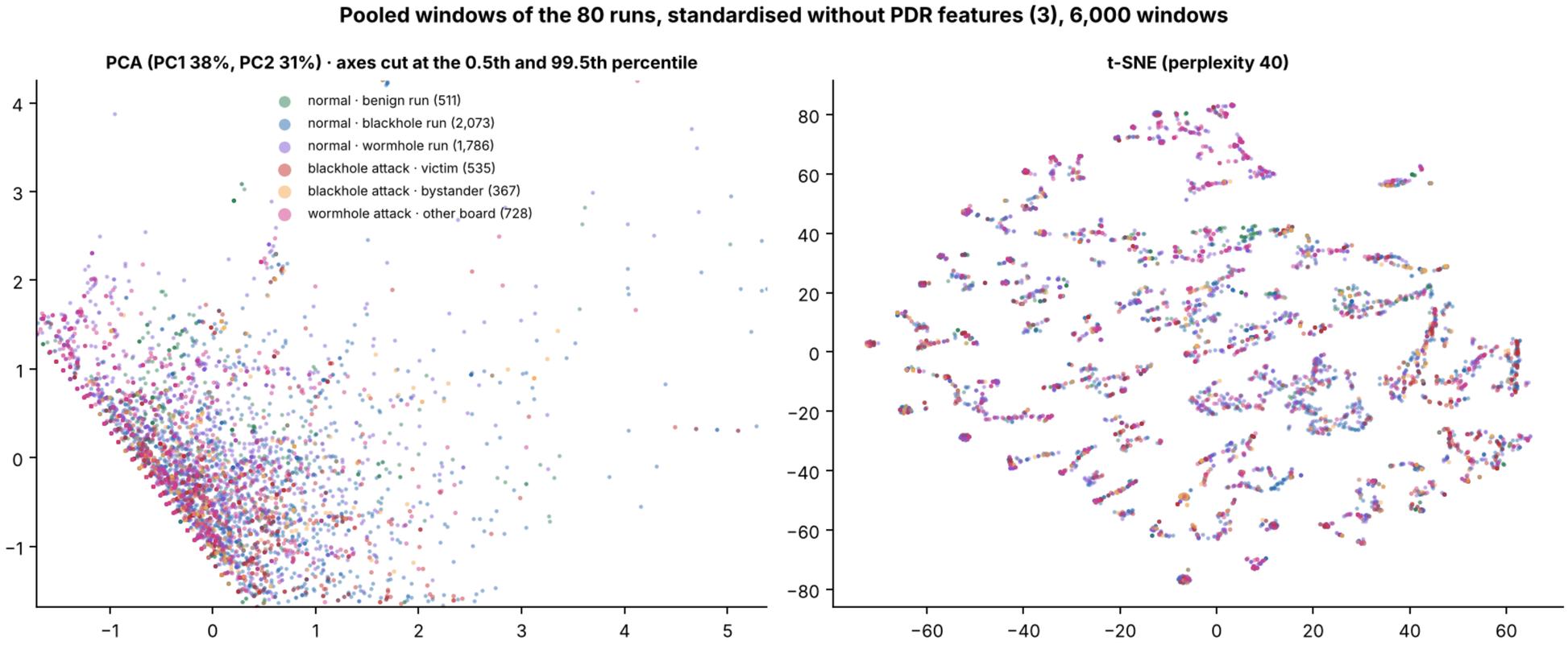
Figure 6.30: pooled 80 runs, FULL set



How to read it

Question: Pooled 80 runs, all 8 features: does any group stand apart in 2-D?
How to read: Each dot = one second (6,000 sampled). Left: PCA (30% and 19% of the spread, extreme dots cut off). Right: t-SNE (only nearness counts). Six colours = three kinds of normal and three kinds of attack second.
What you see: Only the red dots (blackhole victims under attack) stand apart, in both pictures. The other five groups, wormhole attack included, are mixed together.

Figure 6.31: pooled 80 runs, PDR removed



How to read it

Question: The same pooled picture with PDR removed: anything left?
How to read: Only the three RSSI features are used. Left: PCA (38% and 31% of the spread). Right: t-SNE. Same six colours as Figure 6.30.
What you see: All six groups are mixed, the red one too: without PDR the pooled data show no structure.

Compared with published studies

Measure	This study	Published	Reading
Baseline PDR	0.998 victims; 0.9994–1.0000 benign	> 0.97, Khan et al. [22], ESP32	Agree in size; counting differs
PDR under blackhole	0.000	25–40% delivery, Airehrour et al. [70], RPL	Same direction; larger here (attacker on the path by placement)
Parent changes under blackhole	0.00000 → 0.00009 per s	16 to 246 rank changes [70]	Differs: their attack acts on routing (D-20)
Wormhole hop count	Constant	4 → 3 hops, Zhukabayeva et al. [71], ZigBee	Their tunnel is a route; ours an app relay
Wormhole delay	3.9 ms spread; B's LatencyHopRatio unchanged	80–100 → 120–200 ms (about +40%) [71]	Different measure
Duplicate arrivals	99.86% of B's probes; 0 in baseline	Proposed as a signature, Ramírez Gómez et al. [72]	Same evidence, different layer

Platforms, scales and protocols differ, so the comparison is about direction and rough size, not equal values.

7

Chapter 7: Conclusion

What the objectives came to, what we contribute, our limits and recommendations

The three objectives, answered

1	**Build an ESP32 mesh testbed with normal and attack traffic**	**Achieved**	80 of 80 runs; attacks confirmed in 72 of 72, benign in 8 of 8. Qualifiers: emulated attacks; partial mesh = branched tree
2	**Collect PHY, MAC and network features into a labelled dataset**	**Achieved**	16 features, labels from the schedule, every change documented (D-1 to D-37). Qualifier: no MAC retry count (D-15)
3	**Evaluate separability with EDA and clustering**	**Differs by attack**	Blackhole: through PDR only (0.514 vs 0.515). Wormhole: not with cross-layer features (−0.012 to 0.009)

The expected cross-layer evidence was not found, because both attacks left radio and routing unchanged. We report this as a finding, not a failure.

Limits and what should come next

Limitations

- Application-layer attacks only
- Only 8 boards, three campus sites (one semi-outdoor)
- No direct MAC retry count; sniffer on some runs only
- One run per condition
- Strong results rest on features tied to how the attacks were built

Recommendations

1. **Attack routing directly**
   modifiable routing or forged management frames
2. **Collect more normal data**
   benign runs in every cell, with repeats
3. **Improve network measurement**
   a sniffer on every run
4. **Test more wormhole scenarios**
   tunnel other nodes' traffic, other layouts
5. **Evaluate detectors fairly**
   hold out whole runs; beat a PDR or duplicate baseline

Findings and contributions

Four findings

1. Blackhole: clear delivery loss among victims; PDR is the strongest indicator
2. Wormhole: duplicate arrivals at the root, consistently and only during the attack
3. Retransmissions, parent changes, hop counts and signal-to-hop did not change
4. Clustering depends on the feature tied to the attack: remove it and separation collapses

Three contributions

1. **Testbed**
   An ESP32 mesh for normal operation and two attacks under varied conditions
2. **Dataset**
   80 runs with topology, site, scenario, role and phase on every row
3. **Verification**
   A process that separates real attack effects from missing data or broken runs

Thank you

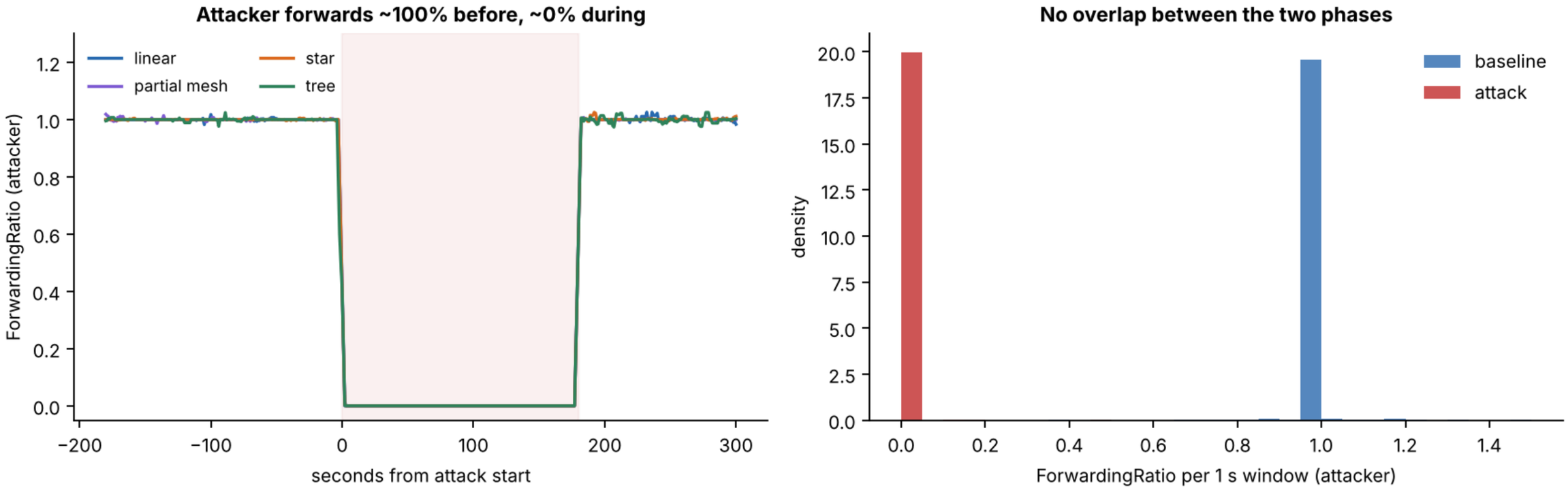
B

Backup: the other figures

B1

Backup B1: blackhole figures

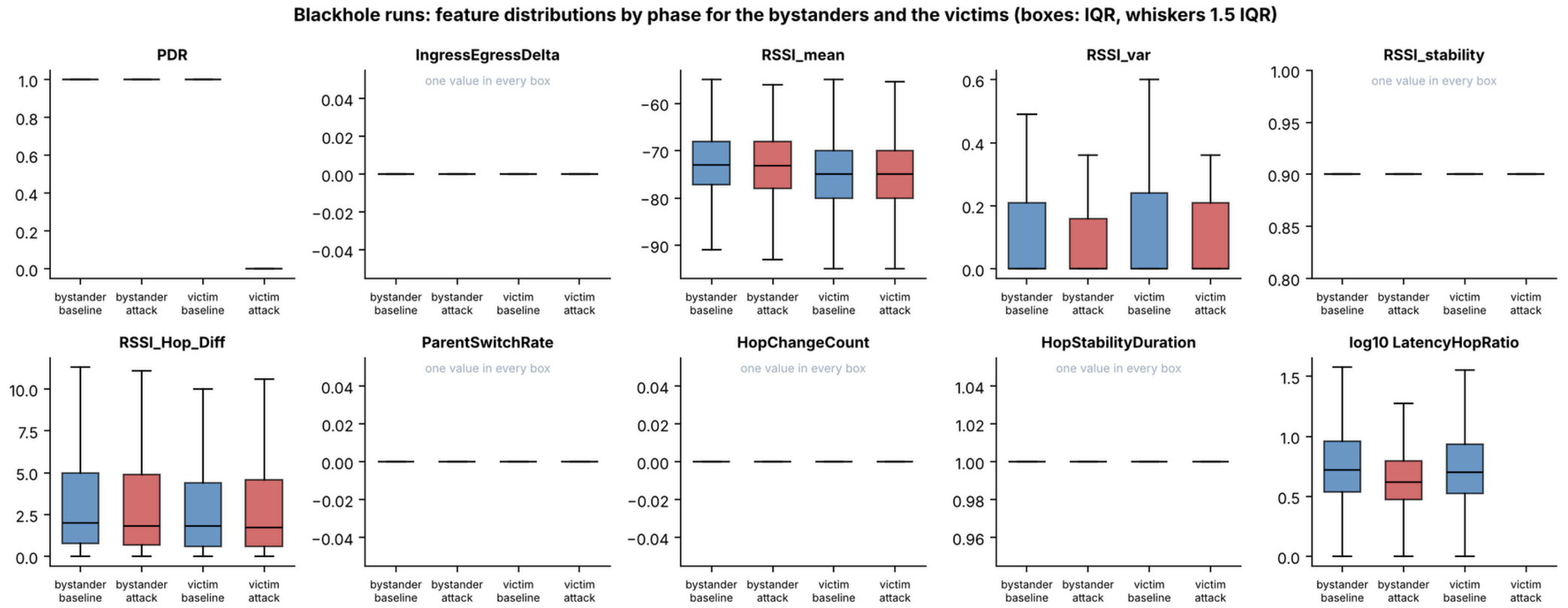
Figure 6.3: the attacker's ForwardingRatio



How to read it

Question: Did the attacker really stop forwarding, and only during the attack?
How to read: Left: time across, ForwardingRatio up (1 = forwards everything, 0 = nothing), one line per topology, shaded = attack. Right: how often each value occurs; blue = baseline, red = attack.
What you see: The line drops from 1 to 0 for exactly the 180 s of the attack and then comes back. On the right, the red and blue bars sit far apart and never overlap.

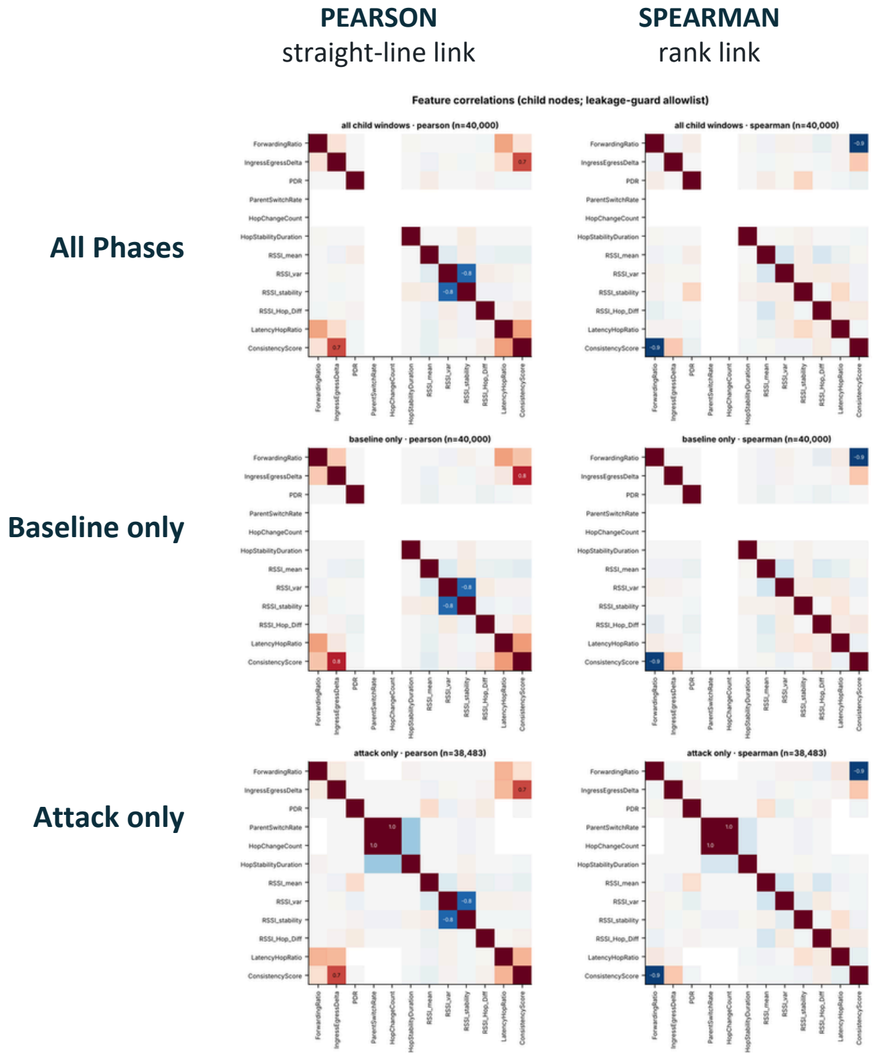
Figure 6.9: blackhole feature distributions by phase



How to read it

Question: Which features change for victims and bystanders when the blackhole starts?
How to read: One panel per feature; four boxes each: bystander and victim, baseline (blue) then attack (red). A box covers the middle half of the values.
What you see: Only PDR moves: the victims' attack box drops to 0. The rest overlap or are one flat value. LatencyHopRatio has no victim attack box: no probe arrived, so there is no delay.

Figure 6.10: correlations between the features



How to read it

What data

Blackhole runs only: the child boards of the 36 blackhole runs (up to 40,000 sampled seconds per square grid). Not the two attacks combined.

How to read

Each square = one pair of features. Its colour is their correlation, from −1 to +1 (scale below). The diagonal is a feature with itself (always +1). Numbers are printed only where the value is 0.5 or more either way. Empty row = the feature never changed.

Left vs. right

Pearson asks whether two features move together along a straight line. Spearman asks only whether one goes up when the other goes up, in any shape. Example: ConsistencyScore vs. ForwardingRatio is 0.12 Pearson but −0.94 Spearman.

Bottom line

Only 4 pairs are strong, and each has a reason in how the features are computed (see notes). No other pair of the 8 clustering features goes past ±0.4.

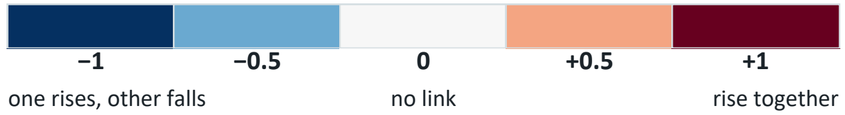
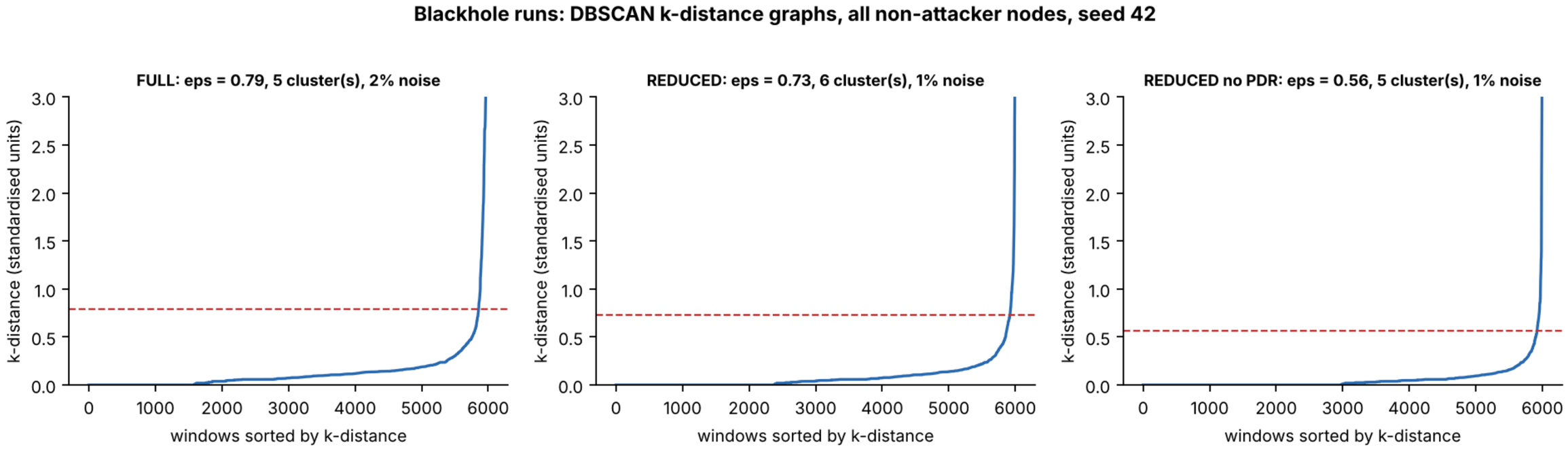
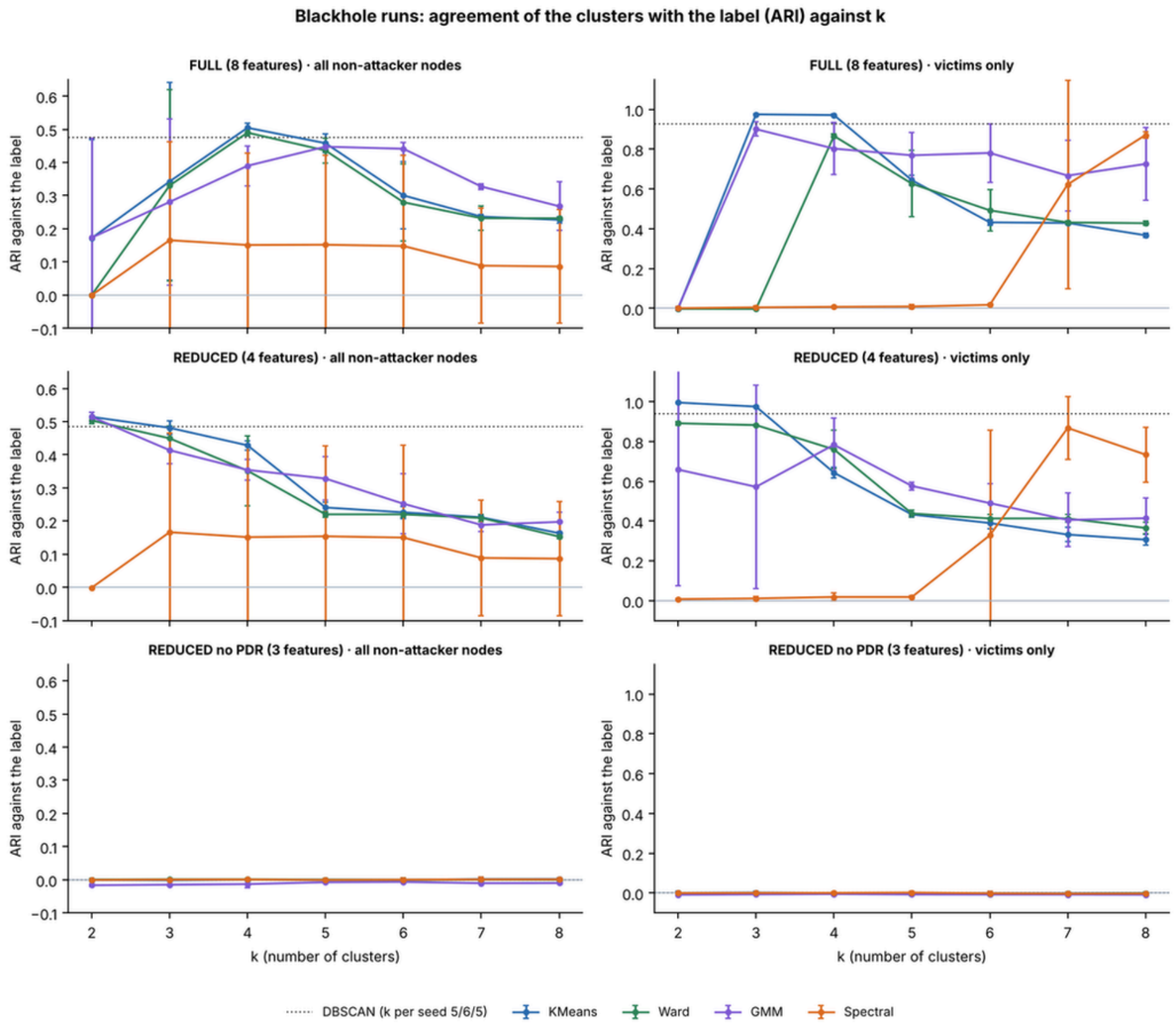


Figure 6.17: choosing eps for DBSCAN



How to read it	**Question:** How did we choose DBSCAN's neighbourhood size (eps)? **How to read:** For each of the 6,000 sampled seconds, take the distance to its k-th nearest neighbour (k = min_samples) and sort from smallest to largest: that is the blue curve (across = rank, not time). Red dashed line = the eps we chose. **What you see:** The long flat part is crowded seconds; the sharp bend is where seconds become isolated. eps is set at the bend: 0.79 (FULL), 0.73 (REDUCED), 0.56 (no PDR). Seconds above the line are noise (1 to 2%).

Figure 6.18: agreement with the label against k



How to read it

The question

How well do the clusters match the real attack label, for each number of clusters k?

How to read

Across = k, the number of clusters asked for (2 to 8). Up = ARI: 1 = perfect match, 0 = no better than chance. Rows = feature sets; left = all non-attacker nodes, right = victims only. One line per method (bars = spread over 3 seeds); dotted = DBSCAN, which picks its own k.

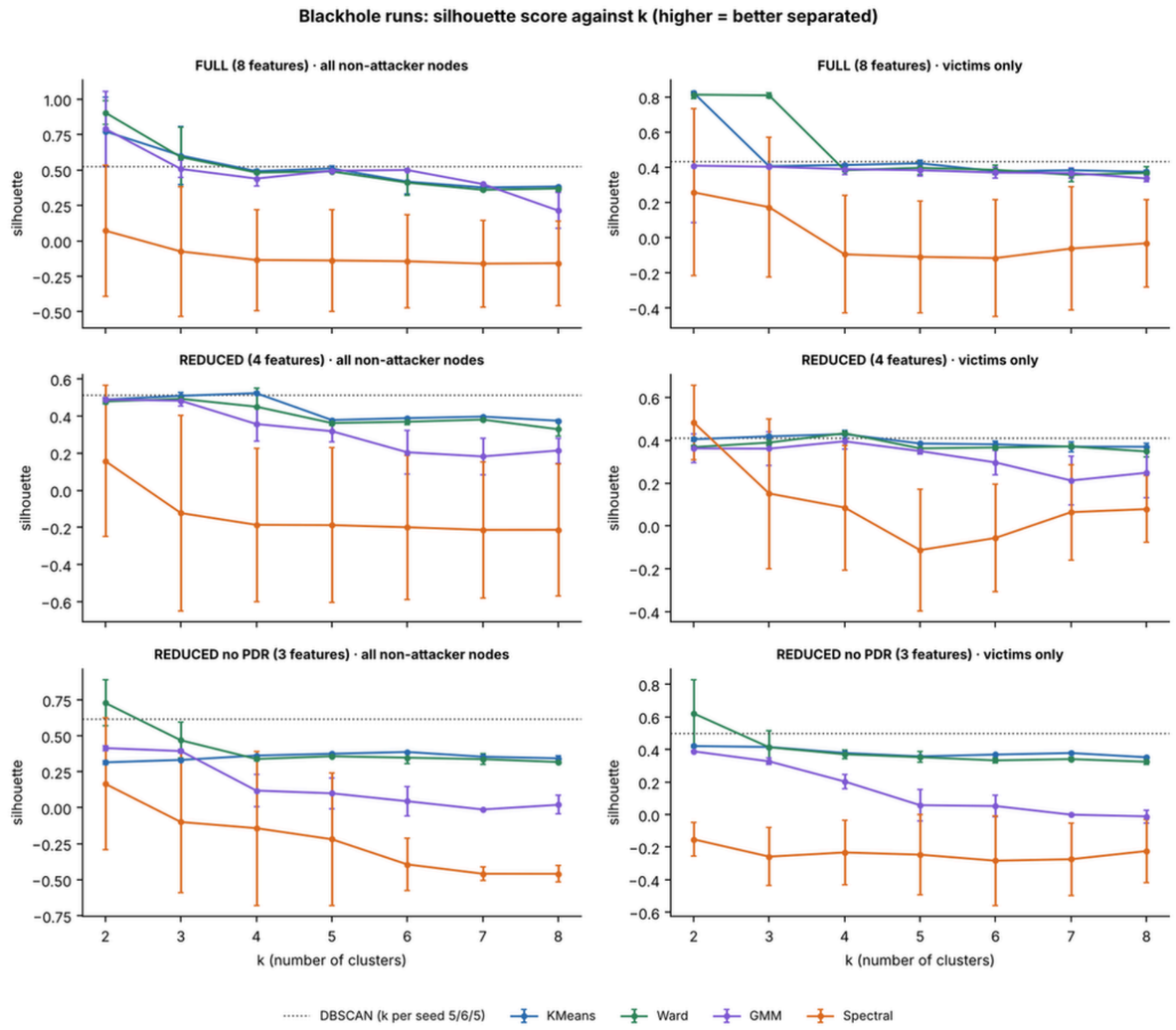
What you see

With PDR (top two rows) the methods find the attack. In the bottom row (PDR removed) every line is flat at 0.

Bottom line

K-Means, REDUCED, k = 2: 0.514 on all nodes (0.515 is the best possible) and 0.996 on victims only.

Figure 6.19: silhouette against k



How to read it

The question

Which number of clusters k gives the cleanest-looking groups?

How to read

Same layout as Figure 6.18. Up = silhouette score: near 1 = tight groups far apart, near 0 = groups that blur together. It judges shape only and never looks at the label.

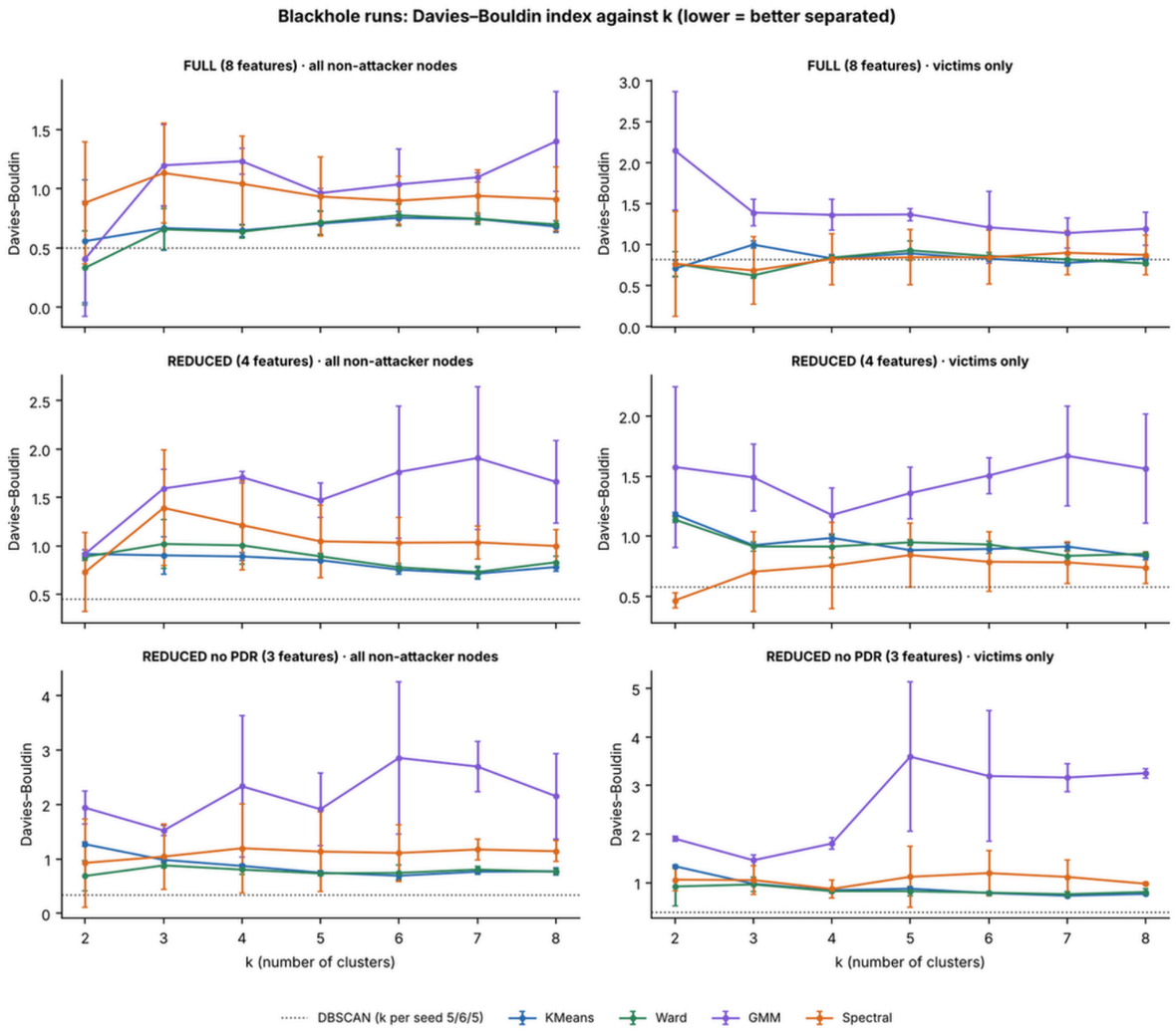
What you see

On FULL (top left), K-Means (0.77) and Ward (0.90) look best at k = 2. But on 2 of the 3 seeds that split only cut off a handful of odd seconds, not the attack.

Bottom line

K-Means ARI at k = 2: only 0.172 (0.515, 0 and 0 over the seeds). The attack appears from k = 4 (0.505). A high silhouette alone can pick the wrong k.

Figure 6.20: Davies–Bouldin index against k



How to read it

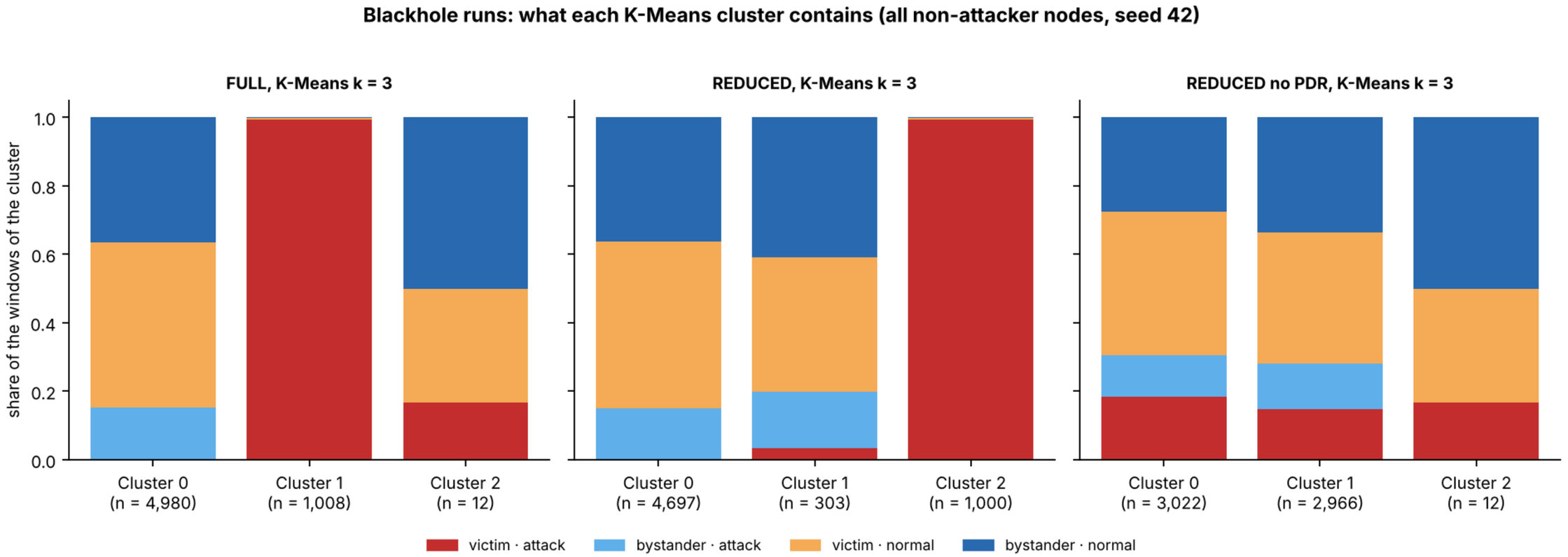
The question
A second shape check: which k separates the groups best?

How to read
Same layout as Figure 6.18. Up = Davies–Bouldin index: lower = tighter, better-separated groups (0 is ideal). Like the silhouette, it never looks at the label.

What you see
No single k is best for every method and feature set.

Bottom line
So k was chosen from the elbow, the silhouette and what the clusters mean, together.

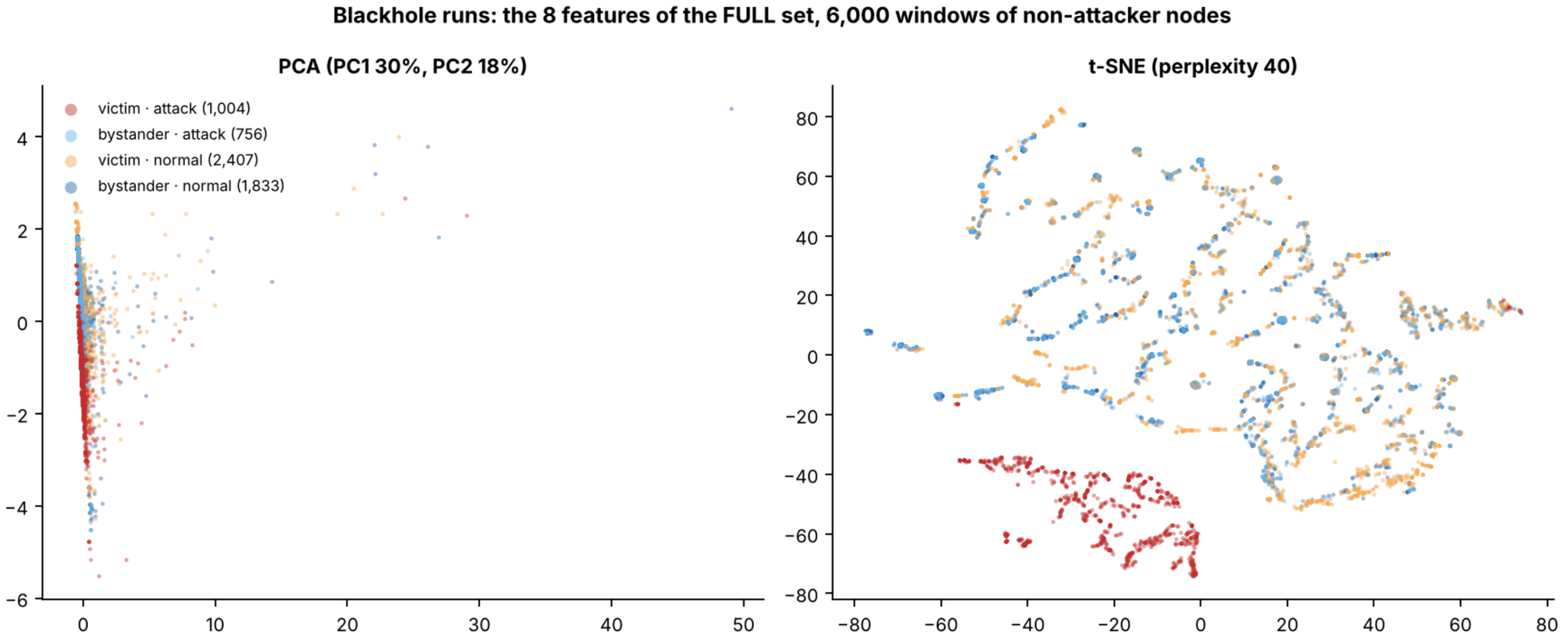
Figure 6.21: what each K-Means cluster contains



How to read it

Question: What is inside each K-Means cluster (k = 3)?
How to read: Each bar = one cluster (n = its size), split by kind of second: red = victim attack, light blue = bystander attack, orange = victim normal, dark blue = bystander normal.
What you see: With PDR (left, middle) one cluster of about 1,000 seconds is almost all red: the victims under attack. Without PDR (right) every cluster is a mix.

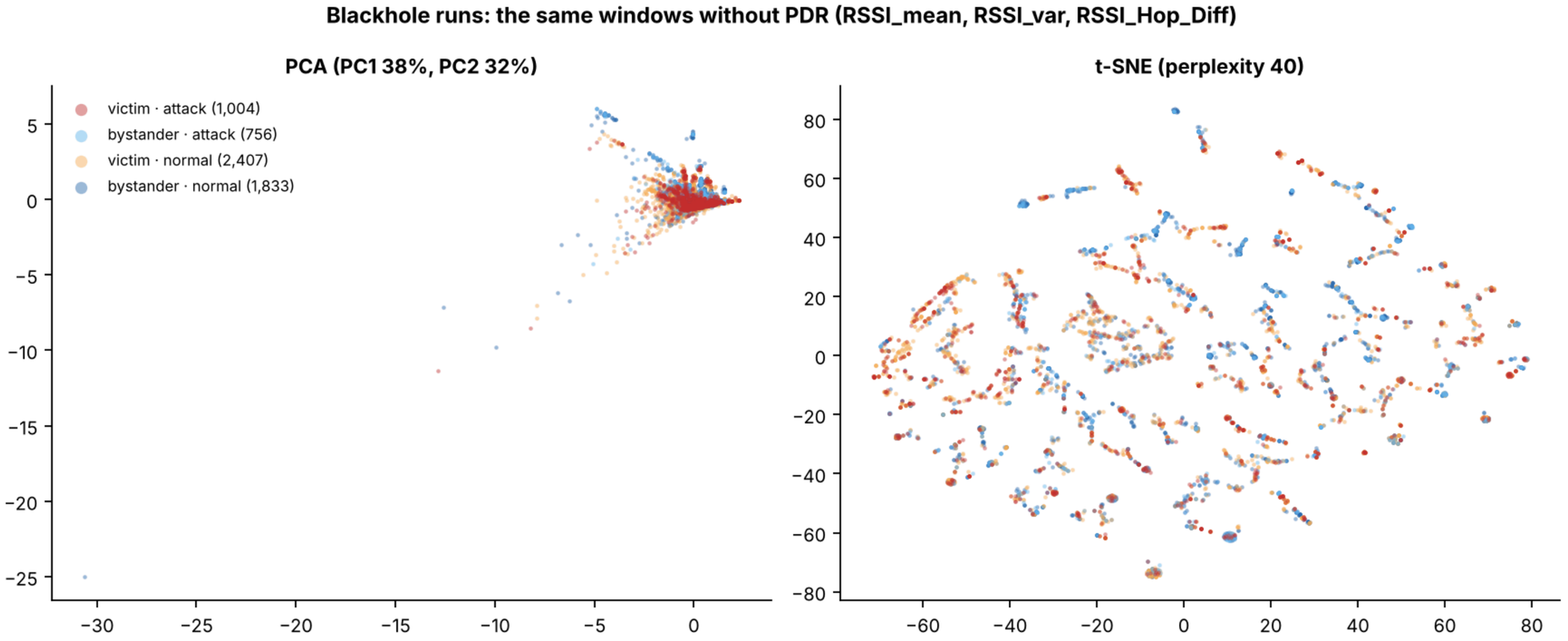
Figure 6.22: PCA and t-SNE, FULL set (blackhole)



How to read it

Question: If we flatten the 8 features into a 2-D picture, do the attack seconds stand apart?
How to read: Each dot = one second of one board. Left: PCA, the two directions with the most spread (30% and 18%). Right: t-SNE, close dots are similar, the axes have no units. Colours as in Figure 6.21.
What you see: In the t-SNE the red dots (victims under attack) form their own island at the bottom. The PCA is squeezed into a thin column by a few seconds with extreme RSSI variance.

Figure 6.23: the same windows without PDR



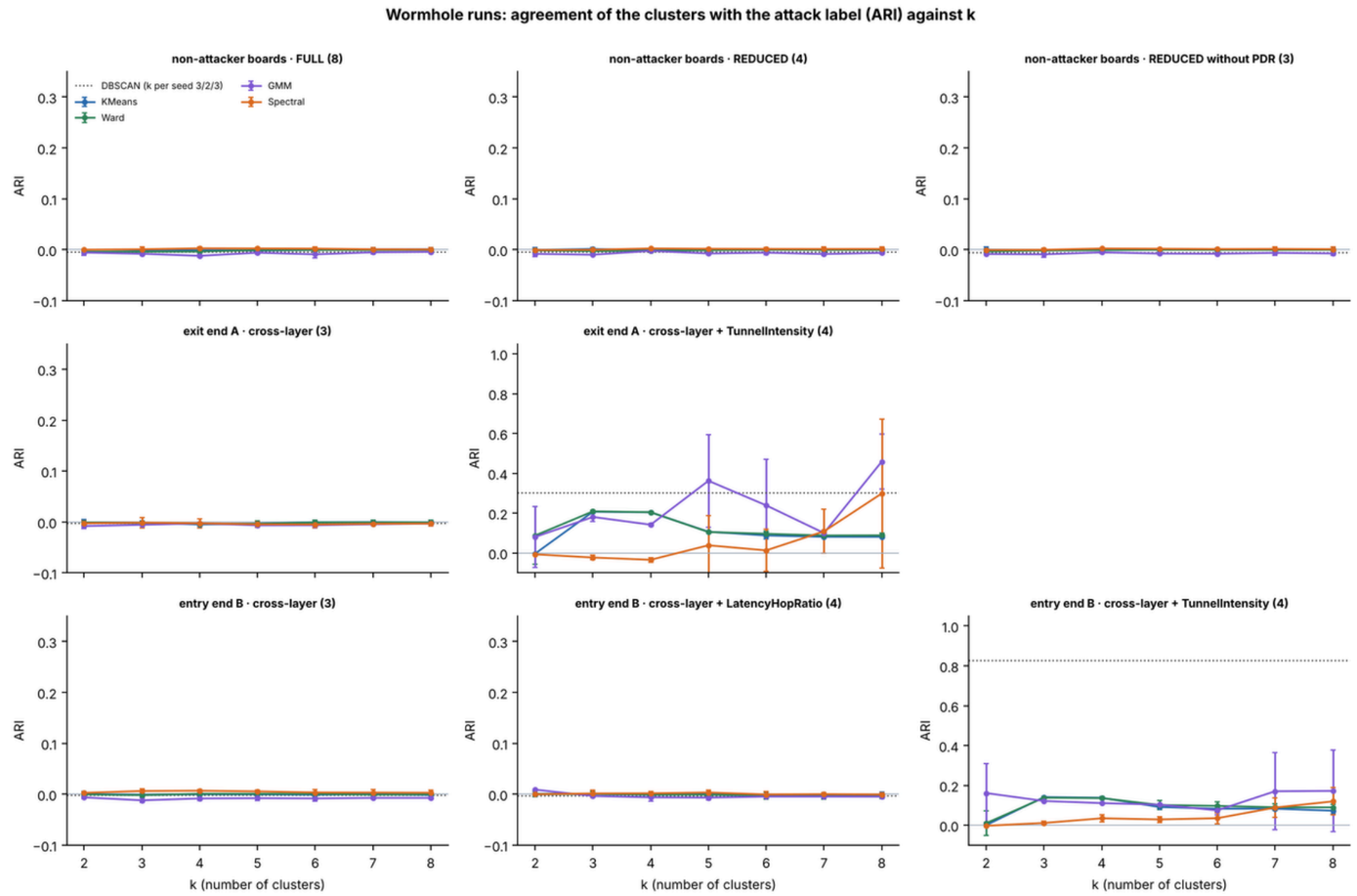
How to read it

Question: The same picture with PDR removed: is anything left?
How to read: Only the three RSSI features are used. Left: PCA (38% and 32% of the spread). Right: t-SNE. Same colours as Figure 6.22.
What you see: All four colours lie on top of each other: without PDR there is no attack group to find.

B2

Backup B2: wormhole figures

Figure 6.24: wormhole clustering against k



How to read it

The question

Can clustering find the wormhole, and on which boards?

How to read

Each panel = one group of boards and one feature set (number of features in brackets). Across = k; up = ARI (0 = chance, 1 = perfect). Top: non-attacker boards; middle: exit end A; bottom: entry end B. Only the TunnelIntensity panels go up to 1.0.

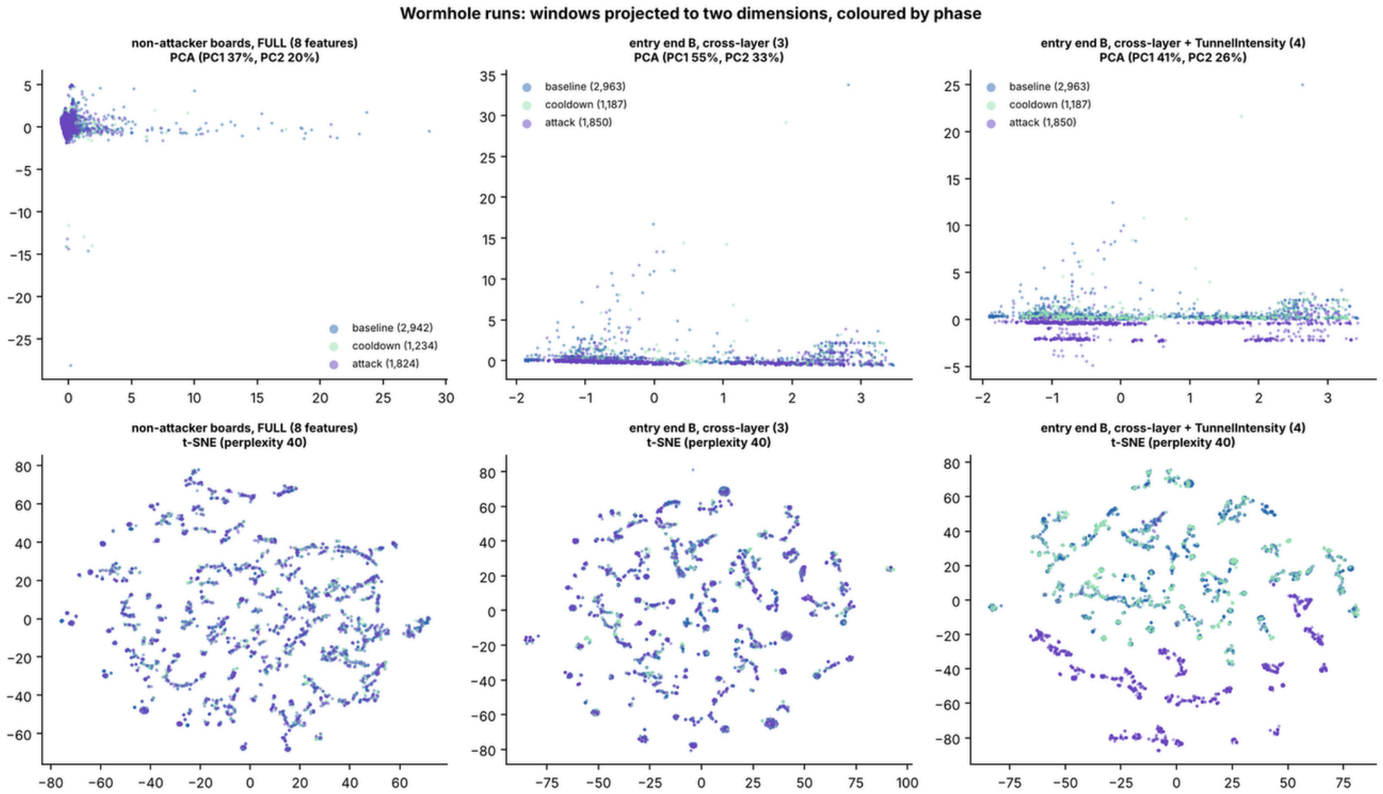
What you see

Every line is flat at 0 for the non-attacker boards and for the cross-layer features alone. A score appears only once TunnelIntensity, the tunnel's own counter, is added.

Bottom line

DBSCAN on end B with TunnelIntensity: ARI 0.827.

Figure 6.25: PCA and t-SNE of the wormhole runs



How to read it

The question

Do the wormhole attack seconds stand apart in a 2-D picture?

How to read

Top row = PCA, bottom row = t-SNE. Columns: non-attacker boards; end B with cross-layer features; end B with TunnelIntensity added. Purple = attack, blue = baseline, light green = cooldown.

What you see

In the first two columns the colours are fully mixed. Only in the third do the purple dots split off, because TunnelIntensity is 0 in every normal second.

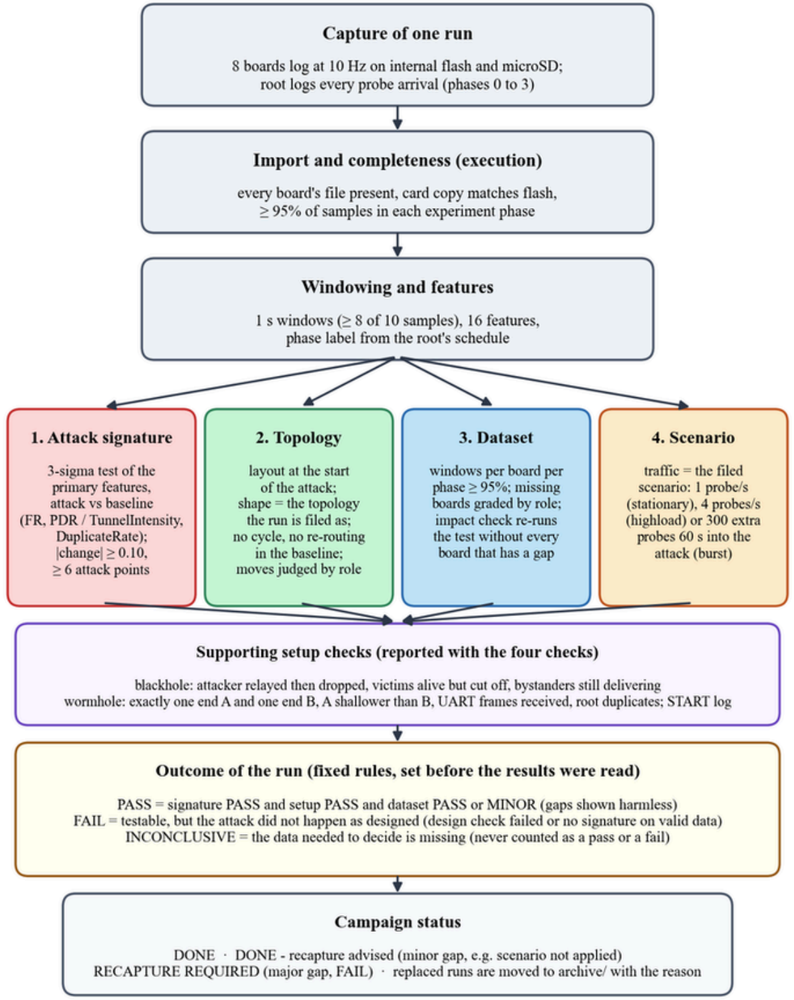
Bottom line

The wormhole shows only in the tunnel's own counter, not in the cross-layer features.

B3

Backup B3: shared figures

Figure 4.55: the order of the verifier's checks



How to read it

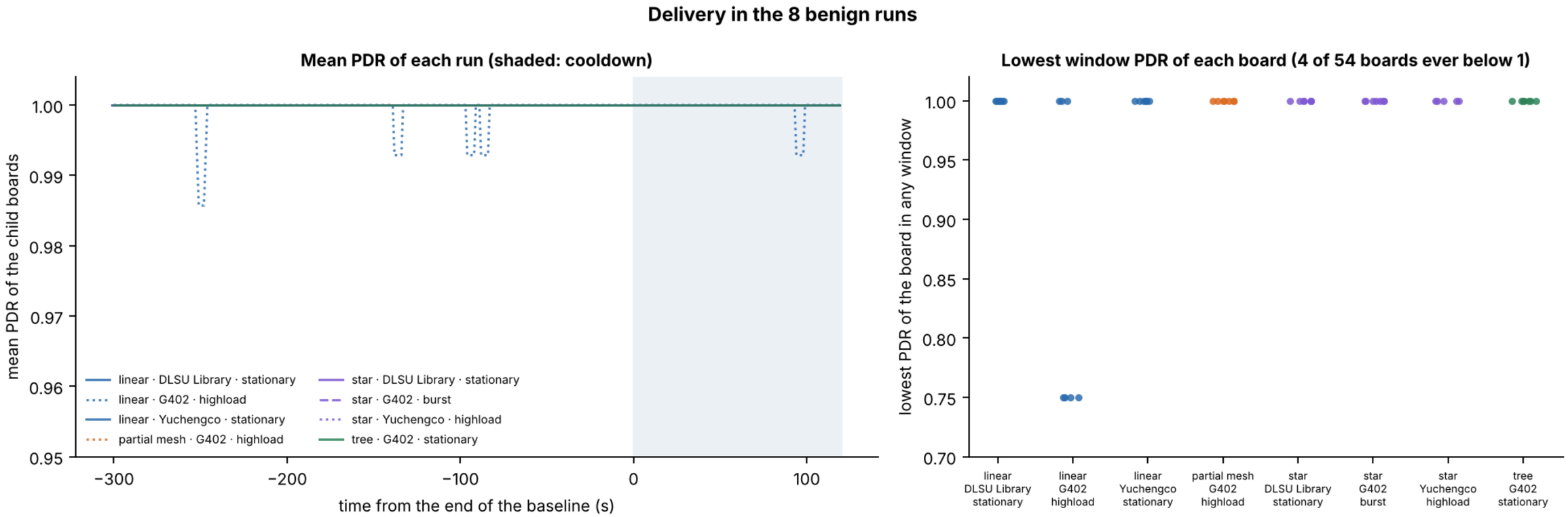
The question
In what order does the verifier check a run before it enters the dataset?

How to read
A flowchart, read top to bottom. Grey boxes: capture, import, features. Then four checks side by side: red = attack signature, green = topology, blue = dataset, orange = scenario. Purple = setup checks; yellow = how the outcome is decided.

What you see
Every run goes through the same path, and the outcome rules were fixed before any result was read.

Bottom line
Three outcomes: PASS, FAIL or INCONCLUSIVE. Missing data is never counted as a pass or a fail.

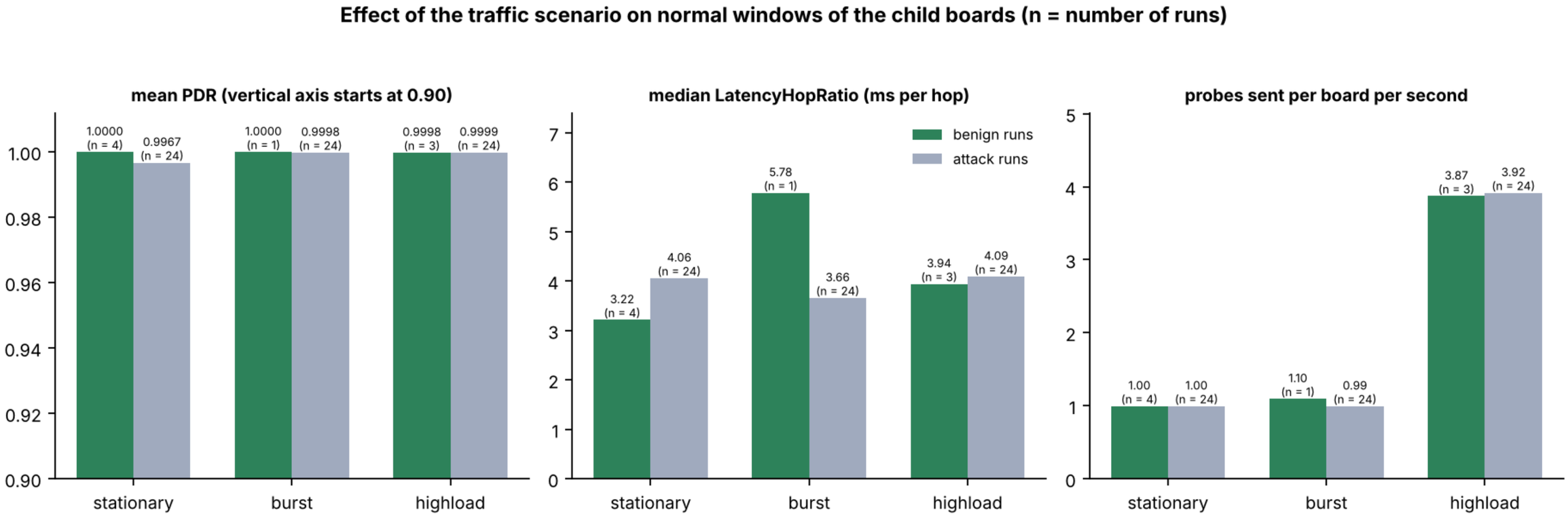
Figure 6.13: delivery in the 8 benign runs



How to read it

Question: Did delivery stay healthy in the 8 runs without any attack?
How to read: Left: each run's average PDR over time (0 = end of baseline, shaded = cooldown); the axis starts at 0.95 so tiny dips show. Right: each dot = one board's worst second.
What you see: All runs stay at 1.00 apart from small dips in one run. Only 4 of 54 boards ever dropped below 1, all in linear / G402 / highload (lowest 0.75).

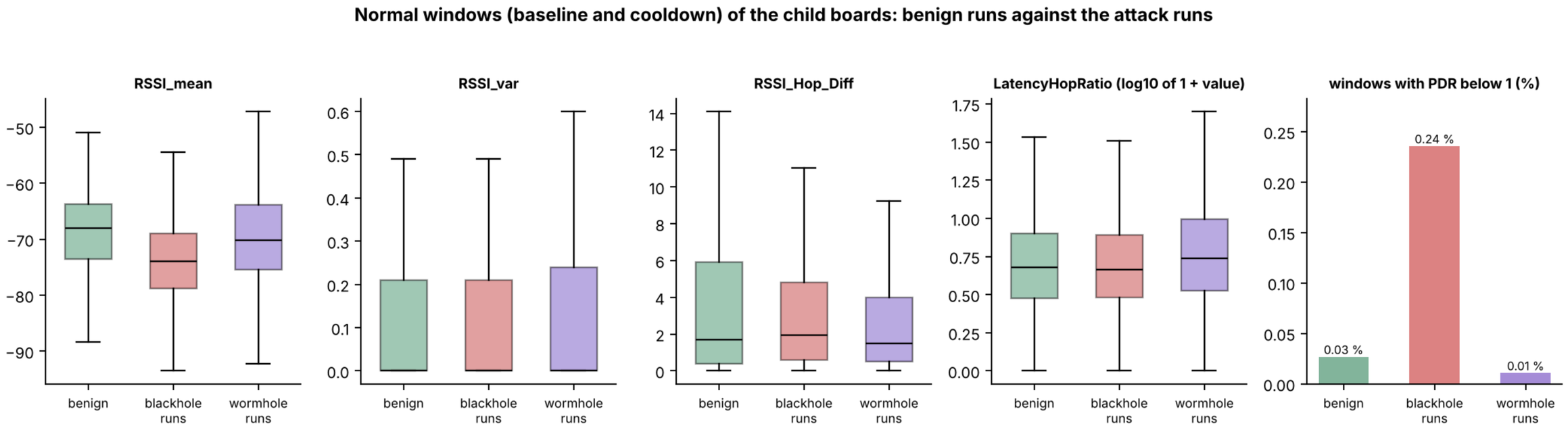
Figure 6.14: effect of the traffic scenario



How to read it

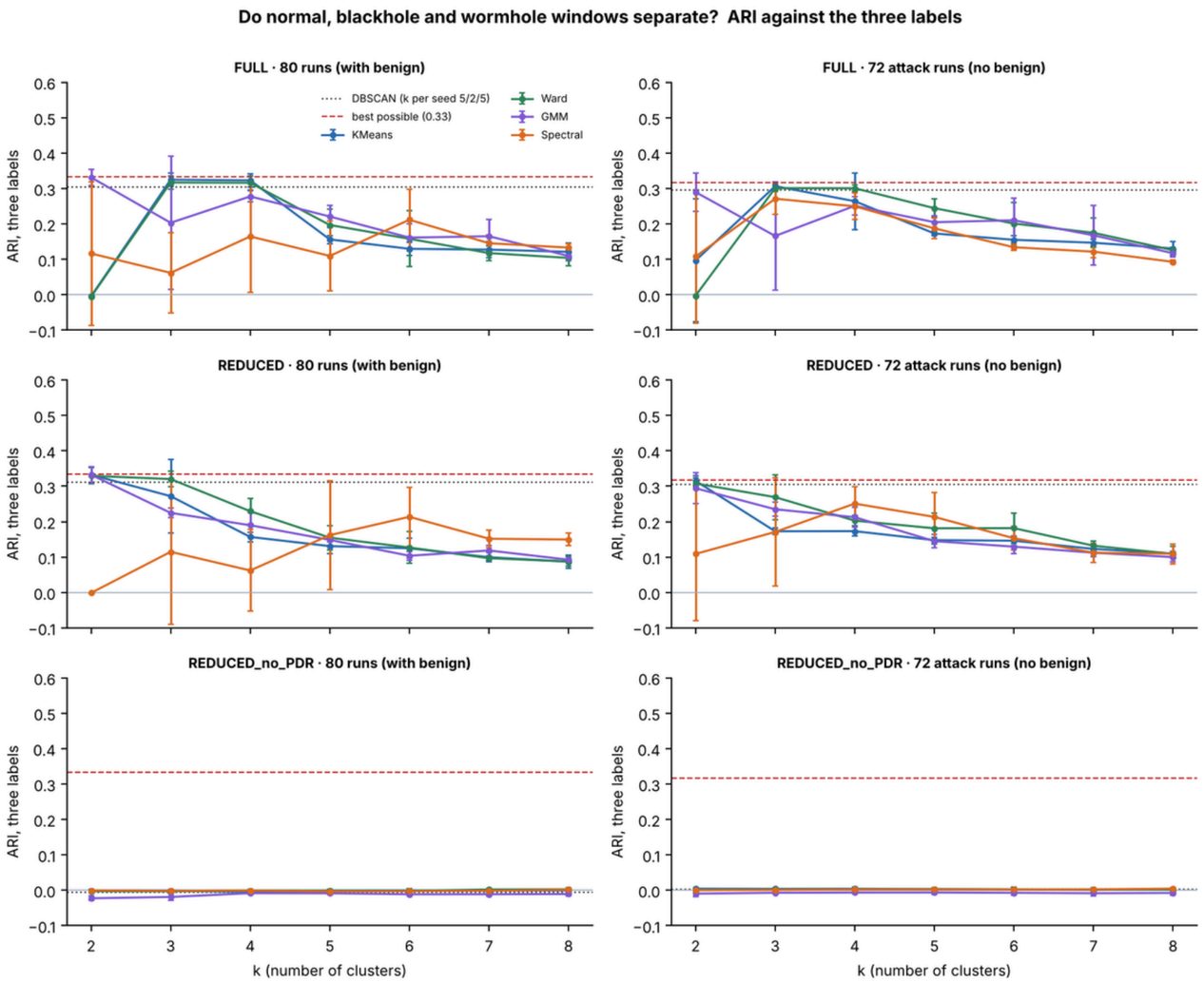
Question: Does each traffic scenario behave the same in benign and attack runs?
How to read: Three panels: average PDR (axis starts at 0.90), typical delay per hop (ms), probes sent per board per second. Green = benign runs, grey = normal seconds of attack runs; n above a bar = runs.
What you see: Probe rate and PDR match, so a scenario means the same in both kinds of run (highload = about 4 probes/s). Delay per hop differs most in burst, which has only 1 benign run.

Figure 6.15: normal windows, benign vs. attack runs



How to read it	**Question:** Are the normal seconds of benign runs like the normal seconds of attack runs? **How to read:** One panel per feature, baseline and cooldown only: green = benign runs, red = blackhole runs, purple = wormhole runs. Box = middle half. Last panel = % of seconds with PDR below 1. **What you see:** Only RSSI_mean differs clearly (the blackhole runs are lower). That comes from the set-up of each capture, not from an attack (see Figure 6.16).

Figure 6.26: normal, blackhole and wormhole windows



How to read it

The question

Can clustering tell normal, blackhole and wormhole seconds apart when all runs are pooled?

How to read

Across = k; up = ARI against the three labels. Rows = feature sets; left = all 80 runs, right = the 72 attack runs. Red dashed = the best possible score (0.33 and 0.32): finding exactly the victim seconds and nothing else.

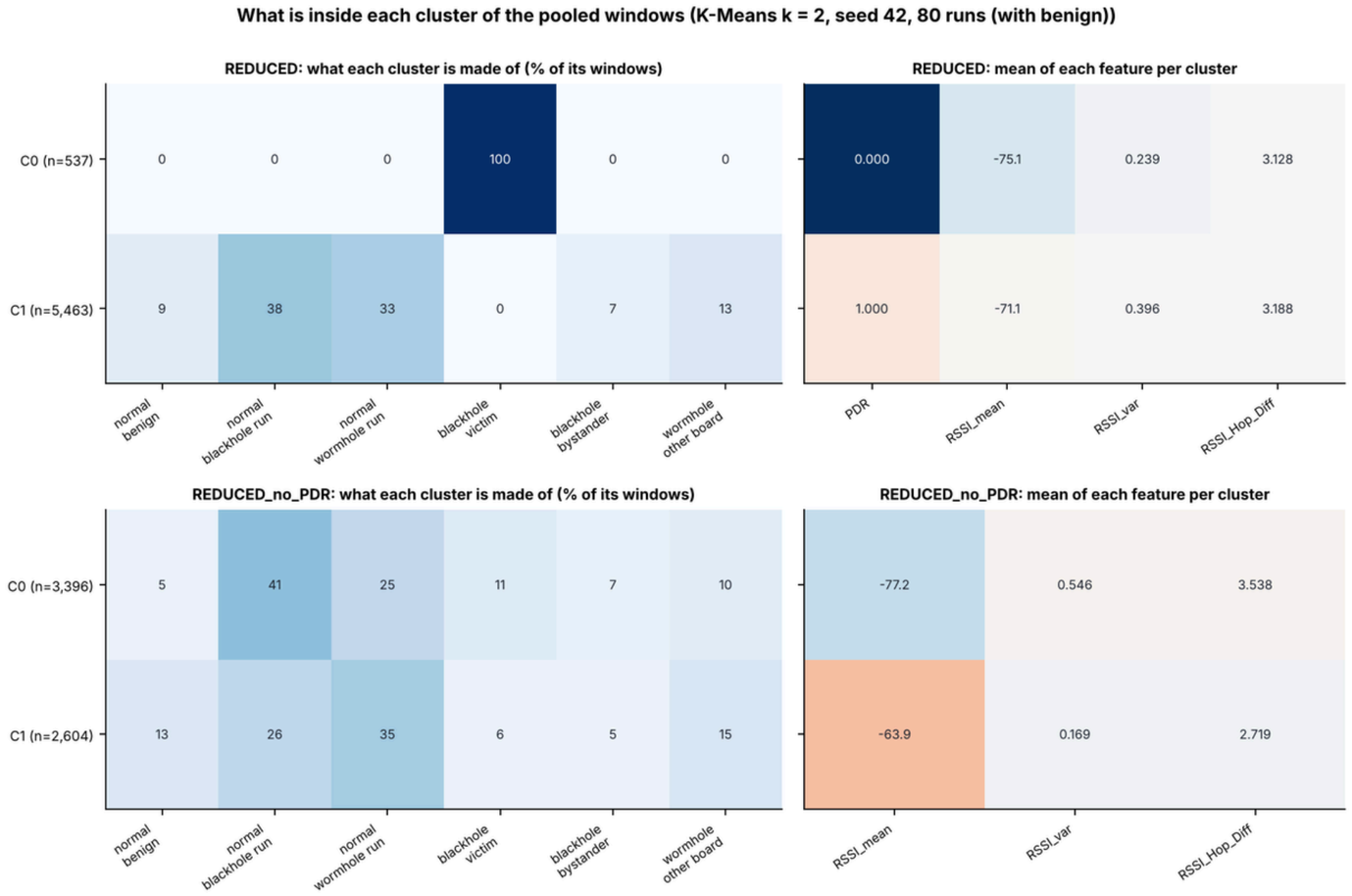
What you see

K-Means reaches the red line (k = 2 on REDUCED, k = 3 on FULL) but never passes it, so the wormhole seconds are never found. Without PDR every line is at 0.

Bottom line

REDUCED, 80 runs: K-Means 0.333, best possible 0.334.

Figure 6.27: pooled clusters, K-Means k = 2



How to read it

The question

What is inside the two K-Means clusters of the pooled 80 runs?

How to read

Each row = a cluster (n = size). Left tiles: % of the cluster from each kind of second (darker = more). Right tiles: the cluster's average of each feature. Top = REDUCED, bottom = REDUCED without PDR.

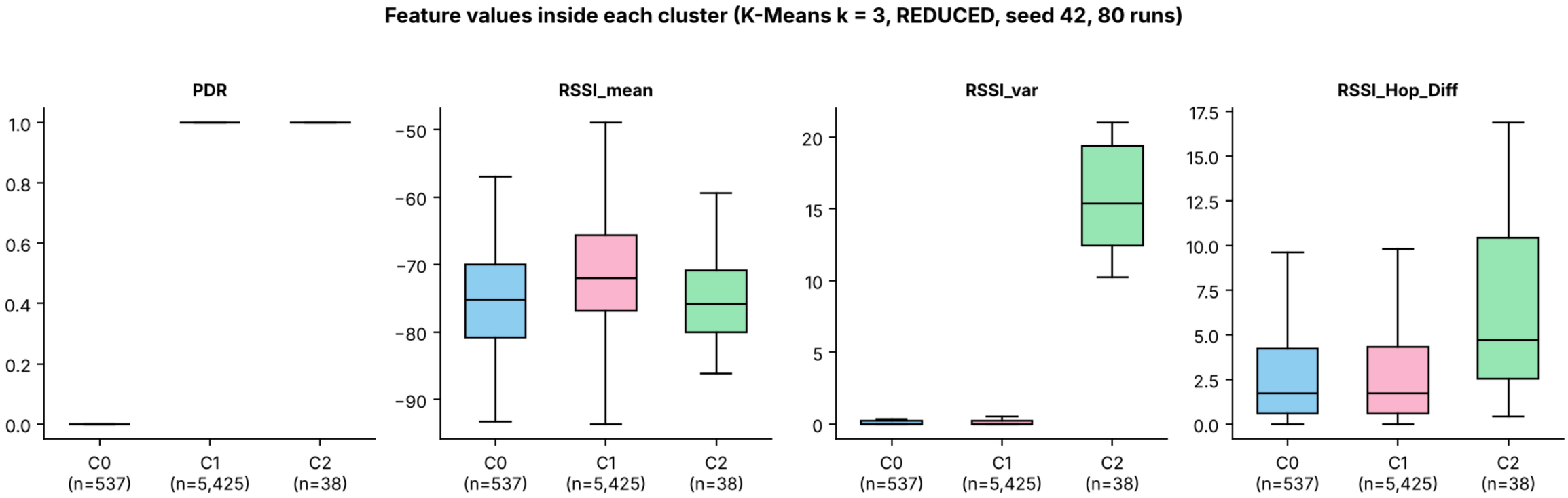
What you see

Top: cluster C0 (537 seconds) is 100% blackhole victims, with PDR 0.000. Bottom: without PDR both clusters mix all six kinds and differ only in signal level (−77.2 vs. −63.9 dBm).

Bottom line

No benign second falls in the victim cluster: it picks up the delivery collapse, not the traffic.

Figure 6.29: feature values inside each cluster



How to read it

Question: What makes each of the three pooled clusters different?
How to read: One panel per feature; one box per cluster (k = 3, REDUCED set) with its size. Blue = C0, pink = C1, green = C2. Box = middle half.
What you see: C0 is defined by PDR = 0 (the victims). C2 is a small group (38 seconds) with very high RSSI variance: noisy seconds, not an attack. The other features overlap.